



MAKING ELECTRONIC TRANSACTIONS **SECURE**

HST Issuer Server API Guide

MISC-HST-ISSUER-SERVER-SPEC
Version 3.12.0 (26.06)

Summary

1. Introduction	4
1.1. Backward Compatibility.....	5
2. Connectivity	6
2.1. URL Scheme	6
2.2. Key Management	7
2.3. Software Architecture and Technology	7
3. Onboarding HST Environment.....	8
3.1. SIK components	8
4. Application Program Interfaces (APIs) – Outbound (From HST to Issuer).....	10
4.1. CheckEligibility	10
4.2. DigitizationNotification	22
4.3. SendPassCode	29
4.4. LifeCycleNotification	35
4.5. DeviceBindingEligibility	40
4.6. ChangeCardInfoNotification	45
4.7. GetSystemStatus	49
4.8. Verify.....	49
5. Tokenization BUS – Inbound (From Issuer to HST)	53
5.1. GetAssociatedTokens	53
5.2. GetTokenInfo	58
5.3. ChangeTokenStatus.....	62
5.4. ActivateToken	66
5.5. GetPANByPANRefID	69
5.6. ChangeCardInfo.....	71
5.7. GetSystemStatus	76
5.8. NotifyResult	76
6. General Objects	79
6.1. CardMetaData.....	79
6.2. TokenInfo	81
6.3. DeviceInfo.....	85
6.4. AuthenticationMethod	87
6.5. ExpirationDate	88
6.6. CardInfo.....	89

6.7. EncryptedPayload	89
6.8. TokenUserInfo	90
6.9. MerchantInfo	91
6.10. RiskInformation	91
6.11. TermsAndConditions	93
6.12. Market	94
6.13. ReturnObject	94
6.14. ErrorObject	94
6.15. ChipData	95
6.16. ChipDataValidationResult	95
6.17. PaymentData	96
6.18. TKData	96
6.19. AuthenticatorInfo	98
6.20. RecentAuthenticationInfo	99
6.21. NotificationResult	99
6.22. ComplementaryAuthenticationInfo	100
7. Return Codes	101
8. Appendices and Supplementary Information	102
8.1. Appendix A	102
8.2. Appendix B	103
9. Revision History	105

1. Introduction

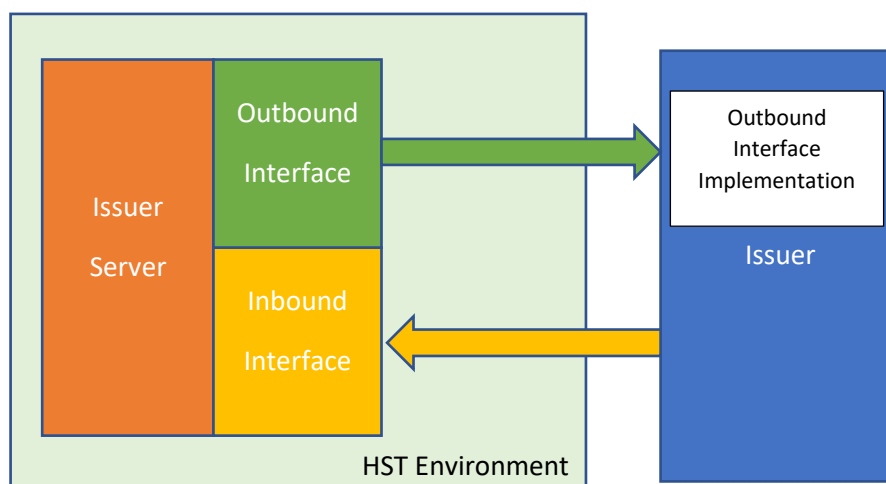
Issuer Server is the system responsible for:

- Authenticating the cardholder during card digitization;
- Defining card metadata during digitization;
- Managing digitized card lifecycle;
- Receiving notifications about digitized cards;
- Providing reports and statistics about the tokenization system;
- Providing detailed information about cards and tokens for support and troubleshooting purposes.

The Issuer Server connects to Visa VTS, Mastercard MDES, American Express and PL Vaults on behalf of the issuers. It also provides an Inbound and an Outbound interface to each Issuer connected to the ecosystem.

The Inbound interface allows Issuers to send life cycle commands to manage digitized cards and to inquire into the system about cards and tokens.

The Outbound interface is used to define cardholder authentication during card digitization and to notify Issuers about token status changes.



Outbound interface implemented by Issuer

1.1. Backward Compatibility

HST ensures that, whenever it is possible, changes to APIs are backward-compatible. The purpose of backward compatibility is to ensure that an API change is seamless and will not impact its utilization in the Issuer environment, in the same way the brands (Visa, Mastercard and American Express) promote updates on their environments for such existent's APIs, guaranteeing the minimal impact possible.

The following changes are considered as backward compatible:

- Adding a new API;
- Adding a new optional request or response element parameter to an existing API;
- Adding a new Enum value;
- Changing the order in which parameters are returned in existing APIs responses.

And for the scenarios above, the Issuer must continue accepting requests and not consider error when a new field is included.

2. Connectivity

The inbound and outbound APIs are designed as RPC style stateless webservice where each API endpoint represents an operation service published that only can be performed using **JSON** payload format. All strings in request and response are UTF-8 encoded and may have a version number API, which allows multiple versions of concurrent APIs to be deployed simultaneously.

Table 01 defines the supported HTTP response codes.

Error Code	Description
200	Success
400	Invalid request
401	Request Denied
403	Not allowed
404	Not found
500	Internal server error
501	Not implemented
503	Service not available

Table 1 – HTTP Response code

2.1. URL Scheme

The URL API follows the scheme below:
scheme://host[:port]/version/apiName

URL ELEMENT	Definition
Scheme	HTTPS
host[:port]	Described in the sections below
Version	v3
API	

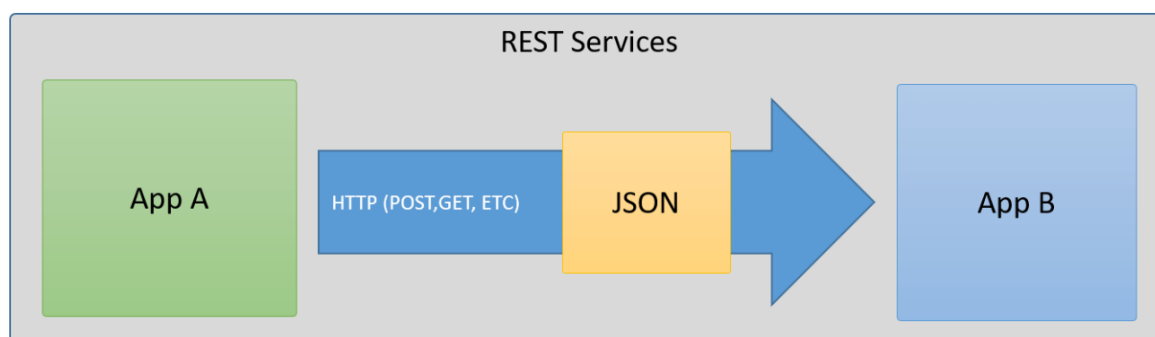
2.2. Key Management

The process of exchanging client/server certificates for the establishment of mutual authentication in TLS 1.2 will be performed by HST (Compliance) and Issuers during the project initial steps.

There is a specific procedure to follow to initiate the certificates exchange process that will be shared with the contact of the Issuer responsible. All the communication will be performed using the kms@hst.com.br e-mail.

2.3. Software Architecture and Technology

The inbound and outbound APIs must be implemented/invoked using **REST API JSON** style.



Implementation using SOAP (XML schemas) **MUST NOT** be used.

3. Onboarding HST Environment

Definition of Parameters

- **Financial Institution Code:** Unique Code defined by HST during Issuer Onboarding that identifies the Issuer at HST Pay Token Services and is out of the scope of this document.
- **Sensitive Information Key (SIK):** It is an AES key generated by HST in its HSM during onboarding and shared between Issuer and HST through kms@hst.com.br e-mail explaining the process.

Notes:

1. Information about the SIK used in testing environment and other dynamic parameters for each issuer will be provided in a specific document.
2. The EncryptedData used in the JSON examples provided in this document were ciphered using the following testing SIKs:
 - **AES-128 key type:** "404142434445464748494A4B4C4D4E4F";
 - **AES-256 key type:** "404142434445464748494A4B4C4D4E4F4F4E4D4C4B4A49484746454443424140".

3.1. SIK components

For **AES-256** the issuer could combine three components by logical XOR operation.

Component 1:

E0E3A481C2E3D1E88E93773B6961B25FCE3A32E23BB0A042075DE2E9E9F15D61

Key Check Value (KCV): CF842B

XOR

Component 2:

B9E81FBDA791DE3AD18AB72F1CE2FB5F4B3C558777659B35BA5F32A49BE86FCC

KCV: ED0AE3

XOR

Component 3:

194AF97F2137499517508A5F39CE074FCA482A29079F723FFA449509315B73ED

KCV: AED09F

SIK (AES 256):

404142434445464748494A4B4C4D4E4F4F4E4D4C4B4A49484746454443424140

KCV (AES): 05E63C

For **AES-128** the issuer could combine three components by logical XOR operation.

Component 1: E0E3A481C2E3D1E88E93773B6961B25F

KCV: A2114B

XOR

Component 2: B9E81FBDA791DE3AD18AB72F1CE2FB5F

KCV: DCC7E1

XOR

Component 3: 194AF97F2137499517508A5F39CE074F
KCV: 900959
SIK (AES 128): 404142434445464748494A4B4C4D4E4F
KCV: 189956

4. Application Program Interfaces (APIs) – Outbound (From HST to Issuer)

The Outbound interface functions are called during card digitization, when an Issuer must be notified about a token status change or to authenticate a user and retrieve available cards associated to the user.

4.1. CheckEligibility

This API is used by Issuer Server to inquire if a card is eligible for digitization. During this process, the card data (Cardholder Name, PAN, CVV and Expiration Date) must be validated by the Issuer. The real PAN must be associated with the TokenRefID or PANRefID elements, because in future calls the actual PAN may not be received. Issuers can **deny** digitization, **approve** it, or approve it with the requirement of additional cardholder identity validation (**ID&V**). In the case Issuer requests ID&V, it must return one or more ID&V methods available for the cardholder.

During the digitization process, there are two final provisioning status that indicates the initial condition of the token when cardholder first tried to provision:

- **Yellow Flow:** tokens that are initially issued in an “inactive” status and are stepped up for ID&V. The issuer must return the value “85” – **card is eligible for digitization and cardholder must be verified** on the *returnCode* element to present to the cardholder the ID&V methods available for identity validation. Cardholders will receive one or more options (*Call Center, App to App and OTP*) depending on Issuer implementation to choose after the card digitization. Cardholders must follow the process till the token activation.
- **Green flow:** tokens issued in an “active” status and no ID&V is performed. Normally applied when the Issuer already has authenticated the cardholder. The issuer must return the value “00” – **card is Eligible for digitization** on the *returnCode* element, and the card is activated right after digitization. For Issuer Wallets who requests cardholder authentication during enrollment or for Merchants which require COF or E-COM Tokens.
The Check Eligibility API also enables the issuer to associate the card being digitized to an internal cardholder identification, typically a bank account or a preexisting user identification.

The identification gives more flexibility to issuers on future calls to identify customers and cards associated with them on HST Environment.

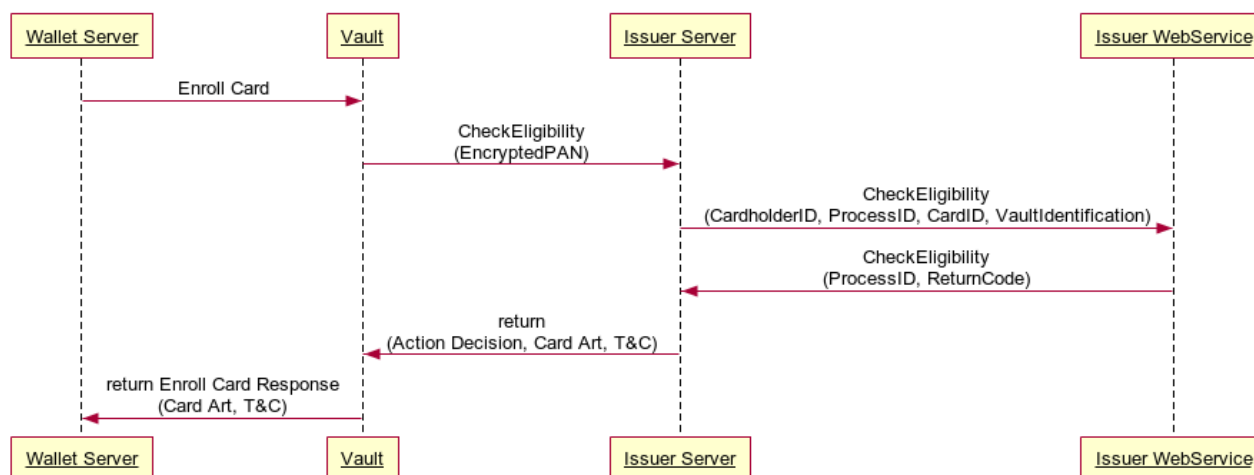
It is recommended to be one of the first APIs to be implemented during an I-TSP/TR-TSP project for a provisioning flow.

Additionally, some Wallet Providers can send, as Token Requestors, cardholder Risk Data to support the Issuers in the decision making for card digitization (green/yellow flow or denial).

This information can optionally be provided in the *riskInformation* element. Depending on the brand, HST can invoke a second (subsequent) call of Check Eligibility API to be able to provide this information to the Issuer.

The expected time for response of this API is approximately 2.0 seconds during the requests are called. Otherwise, the Vault may receive a timeout and the provisioning will fail.

Check Eligibility API



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/checkeligibility	POST
Production: https://{issuer-host:port}/api/v3/checkeligibility	POST

CheckEligibilityRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Size:	64
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32
Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: "VTS" – for Visa; "MDES" – for Mastercard; "AMEX" – for Amex;

	“PL” – for Private Label. Used to identify the Vault in case tokenRefID does not exist in HST database.
Type:	String
Size:	32
Required:	Yes
Element:	walletID
Description:	Identifier of the wallet that generated the request. Some wallet providers such Apple Pay and Google pay define it with the same value used for device identification. For Apple Pay, this field shows the SEID.
Type:	String
Size:	64
Required:	Optional
Example:	Format at HST White Label Wallet: N3GN-KWH6-NTYC-QNKN
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others. <i>Note: Please be aware that VTS will provision the same tokenRefID for a HCE token when a new digitization of the same card occurs within seven days from the token deletion by the cardholder.</i>
Type:	String
Size:	64
Required:	Optional - Present for “VTS” and “MDES” Not present for “AMEX”
Element:	PANRefID

Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated with the Token Requestor, which means it is not unique for a PAN and it can have multiple PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64
Required:	Present for “VTS” Optional for “MDES” Not present for “AMEX”
Element:	encryptedCardInfo
Description:	Encrypted CardInfo. Contains card information to be used on digitization process.
Type:	EncryptedPayload
Required:	Yes
Element:	processID
Description:	Digitalization process identifier generated for each request. For Mastercard, this field must contain the same value during a complete digitization process, and it is sent on the next APIs such as DigitizationNotification and SendPassCode. For Visa and Amex, this value does not remain the same.
Type:	String
Size:	Max 64
Required:	Yes
Element:	userLanguage
Description:	User preferred language according to ISO 639 Version 3 Language Code (for example: “eng”).
Type:	String
Size:	3
Required:	Present for “VTS” Optional for “MDES” Not present for “AMEX”
Element:	Source
Description:	How the card number was obtained. Possible values are: “ON_FILE” – PAN origin is a card number stored in a merchant; “MANUALLY” – PAN was entered by the customer; “MOBILE_APP” – PAN provided by a mobile app. Typically a list of cards provided by the issuer after cardholder authentication;

Type:	"TOKEN" – The source of pan of this token (ECOM or COF) provisioning was issued by a token device bound (NFC/SE). Applicable when wallet has an NFC/SE and it is provisioning a new ECOM/COF token. Also used when NFC/SE token credentials from a device are used for provisioning a new NFC/SE token in other device (Apple Card Continuity feature); "BROWSER" – Indicates that the account details were pulled from a browser for tokenization. "CONTACTLESS_TAP" - PAN was captured using "Tap to Add Card" service
Required:	String
Element:	riskInformation
Description:	Risk data provided by the Wallet Provider or by Vault. This information can help the Issuer in the decision for card eligibility (green/yellow flow or denial). Updated to work with Mastercard Token Provisioning Insights (TPI).
Type:	RiskInformation Object
Required:	Optional
Element:	riskInformationResubmission
Description:	Depending on the brand implementation, Token Requestor risk information can be received after issuer answers to CheckEligibility. In this case, HST invokes the CheckEligibility API a second time to present issuer with this information. If TRUE, this field indicates the call is a resubmission on the API. Depending on Issuer evaluation of the risk data, a different return code can be replied to the second call. On most cases Issuers will switch from a Green flow to a Yellow or Red flow. In case of Yellow Flow, authentication information should be provided on response. The absence of this field means it's the first call to the API.
Type:	Boolean
Required:	Optional
Element:	tokenType
Description:	Information provided by HST using the parameters sent by the Vault to inform the Issuer of the token type requesting the digitization. Possible values are: "HCE", "SE", "COF", "ECOM", "QRCODE".
Type:	String
Required:	Yes
Element:	TokenRequestorName
Description:	Identification of the Token Requestor name requesting digitization. It identifies a Multi Issuer Wallet, an Issuer Wallet or a Merchant, like Uber, Netflix, Adyen, Apple Pay, Samsung Pay and others.
Type:	String
Required:	Optional

Element:	recommendedDecision
Description:	A suggestion provided by HST to support Issuer during the decision flow. This value uses a combination of other values received by the Vault to create a suggested decision. Issuers can use this value to determine a flow to the cardholders. Possible values are: "GREEN", "YELLOW" or "RED".
Type:	String
Required:	Yes
Element:	recommendedDecisionReasonCode
Description:	The code explains the <i>recommendedDecision</i>. Possible values are: <u>Restricted to RED flow recommendation:</u> "0001" – Error due to the card digitized in too many devices. "0002" – Too many consecutive incorrect attempts of digitization (Invalid CVV2 or Expiration Date). "0003" – Token Requestor recommendation. "0005" – Token provisioned abroad. (*) "0006" – Account with suspicious transactions history "0007" – Too many consecutive attempts of token device digitization "0008" – Too many consecutive token device digitization "0009" – Cardholder name mismatches with devices owner. "0010" – Device score is too low "0011" – Account score is too low "0012" – High risk digitization detected <u>Not restricted to the RED flow:</u> "0004" - CVV2 is present for issuer validation. (*) By default, this code recommends RED flow in <i>recommendedDecision</i> element. However, it could be mapped to RecommendedDecision=YELLOW, by opening a support ticket on HST.
Type:	String
Required:	Optional
Element:	recommendedDecisionReasonCodeList
Description:	An array with all the codes reported by the vault and/or by the wallet provider to support the token provision decisioning. The possible values are the same of the listed in the <i>recommendedDecisionReasonCode</i>.
Type:	String[32]
Required:	Optional

Element:	chipData
Description:	Encrypted ChipData Object. This parameter is present only when Issuer supports Tap to Add Card and issuer has decided to receive and validate chip data by themselves.
Type:	EncryptedPayload
Required:	Conditional. MDES only, when <i>Tap to Add Card</i> feature is enabled.
Element:	chipDataValidationResult
Description:	ChipDataValidationResult object, present only when Issuer supports Tap to Add Card and issuer has decided to use Mastercard On-behalf service (OBS) for chip data validation.
Type:	ChipDataValidationResult Object
Required:	Conditional. MDES only, when <i>Tap to Add Card</i> feature is enabled.
Element:	deviceType
Description:	"UNKNOWN", "MOBILE_PHONE", "TABLET", "WATCH", "PC".
Type:	String
Size:	Max 32
Required:	Optional
Element:	deviceInfo
Description:	Information about the device associated with the token.
Type:	DeviceInfo Object
Required:	Optional
Element:	complementaryAuthenticationInfo
Description:	Additional authentication data. It is used in Apple Multi Device Provisioning scenario.
Type:	ComplementaryAuthenticationInfo Object
Required:	Optional
Element:	encryptedTokenInfo
Description:	Encrypted TokenInfo . Contains token information related to this notification.
Type:	EncryptedPayload
Required:	Conditional. MDES only, when source element is " TOKEN ".

CheckEligibilityResponse

Element:	requestID
Description:	Request identifier uniquely generated for each request returned by the Issuer.
Type:	String
Required:	Yes
Element:	processID
Description:	Digitalization process identifier generated for each request returned by the Issuer.
Type:	String
Size:	Max 64
Required:	Yes

Element:	returnCode
Description:	Possible values are: “00” – Card is eligible for digitization; “05” – Card is not eligible for digitization; “85” – Card is eligible for digitization and cardholder must be verified; “16” – Card not found, invalid PAN; “22” – Invalid card security code; “23” – Invalid card Expiration date; “27” – Too many attempts, suspected fraud. Return expected when element "recommendedDecisionReasonCode" value received in request is "0002". For Amex, the following specific digitization denial reasons may be returned by the Issuer: “24”, “25” and “26”. “24” - Card has not been activated, replaced, or renewed card has not been activated; “25” - Non-whitelisted accounts when a market is at beta test phase; “26” - Ineligible instant account/instant membership account provisioning.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional
Element:	cardMetaData
Description:	CardMetaData. This element is not encrypted. In case the Issuer does not send this field, the token requestor will receive the information configured by the issuer in the vault platform (VCMM or Mastercard Connect). For Mastercard, the only field permitted is “productID” (HST Parameter), that must match the “issuerConfigID” (Mastercard Parameter) defined on Mastercard Connect. For Visa, it can be defined by “productID”, that must match the “profileID” (VISA Parameter) defined on the Visa Card Metadata Management Tool (VCMM) or by sending the cardArtID and termsAndConditionsID. For Amex, it is required to return productID, productName and productType elements.
Type:	CardMetaData
Required:	Optional for “VTS” and “MDES” Required for “AMEX”

Element:	authenticationMethods
Description:	Authentication methods list for specific card, if authentication is needed. The possible values for implementation are OTP, Call Center and App-to-App and their details are described on the AuthenticationMethod description element.
Type:	Array < AuthenticationMethod >
Required:	Optional
Element:	userID
Description:	Issuer identification on the cardholder. Typically, an account or online banking user ID. Only for auditing purposes on HST's system, there is no participation during the provisioning and transaction flows.
Type:	String
Required:	Optional
Element:	market
Description:	Market object. Indicates the market/region where the card was issued.
Type:	Market object
Required:	Required only for AMEX.
Element:	expirationDate
Description:	Card expiry date.
Type:	ExpirationDate object
Required:	Required only for AMEX.
Element:	PANSequence
Description:	Funding account PAN sequence. Examples: 00 (Default Value), 01, 02, 03.
Type:	String
Size:	2
Required:	Required only for AMEX.
Element:	chipDataValidationResponse
Description:	The Issuer validated chipData validation result code. Must present if issuer is validating chip data for <i>Tap to Add Card</i> feature. Must be one of the following values: "VALID" = Chip data provided is valid, "INVALID" = Chip data provided is not valid, "NOT_PROCESSED" = Unable to process Chip data validation (issuer temporarily unavailable).
Type:	String
Size:	32
Required:	Conditional, only for MDES.
Element:	virtualCardRelatedData
Description:	Encrypted CardInfo . If the card information received in request refers to a virtual card number (VCN), the Issuer must return in this object the card info related (<u>PAN and expiration date are</u>

Type:	<u>mandatory</u>). These mappings enable generation and assignment by MDES of a consistent PAR (Payment Account Reference) across all VCNs linked to the same card account.
Required:	EncryptedPayload Conditional, used only for MDES if card being tokenized refers to a virtual card.
Element:	merchantAdviceCode
Description:	Indicator to notify the merchant whether the Issuer VCN (Virtual Card Number) is intended for single or multiple use in the payment transaction. Must be one of the following values: "SINGLE_USE" = Consumer single-use virtual card number "MULTI_USE" = Consumer multi-use virtual card number String
Type:	Max 32
Size:	Conditional, used only for MDES if card being tokenized refers to a virtual card.
Required:	
Element:	provisioningDenialReason
Description:	Reason for declining the tokenization request as described below: LOST : The card has been lost. EXPIRED : The card is expired. RESTRICTED : The account or card is currently restricted. SEC_VIOLATION : A security violation has been detected. CRYPTO_FAIL : Cryptogram validation for the account failed. UNACCEPTABLE_PIN : The entered PIN is not acceptable. EXCEEDS_PIN_TRIES : The maximum number of PIN attempts has been exceeded. STOLEN : The card has been reported stolen. INVAL_CARD : The card is invalid. CLOSED_ACCOUNT : The card is closed. INVAL_ACCT : The account is invalid. INVAL_PIN : The PIN entered is invalid. RCVR_NOTPERM : Token Authorization is not permitted to the issuer. SNDR_NOTPERM : The token requestor is not permitted. EXCEEDS_CNT : Validation attempts exceeded. DO_NOT_HONOR : None of the above.
Type:	String
Size:	Max 64
Required:	Optional

JSON Examples

CheckEligibilityRequest

```
{
  "requestID": "2",
  "institutionCode": "HST",
  "vaultIdentification": "VTS",
  "walletID": "N3GN-KWH6-NTYC-QNKN",
  "tokenRequestorID": "42301999123",
  "tokenRefID": "DNITHE381502386342002358",
  "PANRefID": "V-3815023863409817870482",
  "encryptedCardInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "a96b3e84232d573c6592ceda",
    "encryptedData":
"KV1Mgkv40Nt4yggF1Ka7osdIkyMSsVe8K3o9wpQpMRTGeiXV2I65fIYgjZY1IGE pj/A7+KX3XB8C4
Foo8tEZ5xxQXa2PRudQ9B9s9WZbWoANcyADAw7ix7CQUN4x2ps9+oe8UaLtwjKrbKEDFkCML9rE90
oco7vMr7y+uAlZ2NazPoWwx5fcQkn",
    "MACLength": 16
  },
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "source": "MANUALLY",
  "riskInformation": {
    "recommendedDecision": "YELLOW",
    "deviceScore": "2",
    "accountScore": "2"
  },
  "tokenType": "NFC",
  "tokenRequestorName": "HSTPayWallet",
  "recommendedDecision": "YELLOW",
}
```

Where:

//Plain CardInfo Object Data:

```
{
  "PAN": "1111110000000003",
  "expirationDate": {
    "month": "11",
    "year": "2024"
  },
  "CV2": "500",
  "cardholderName": "FRANCISCO PEREIRA"
}
```

CheckEligibilityResponse

```
{
  "requestID": "2",
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "returnCode": "85",
  "cardMetaData": [
    {
      "productID": "14454"
    },
    "authenticationMethods": [
      {
        "identifier": "125485644",
        "type": "bank_app",

```

```

    "maskedInfo": "Mobile Banking App",
    "sourceAddress": "com.DemoBank.DemoApp",
    "platform": "ANDROID"
  },
  {
    "identifier": "125485633",
    "type": "cell_phone",
    "maskedInfo": "XXX-XXX-1234"
  }
],
"userID": "12345678909"
}

```

Card Meta Data Implementation Options

Option 1 (Default):

- The field **cardMetaData** is not sent in the response of this API. The Vault will get the metadata information default loaded on their platform. It is most applicable for scenarios when the Issuer has one card art image for BIN.

Option 2 (for Mastercard and Visa):

- Define a value for the **productID** field for each card product. It is most applicable for scenarios when the Issuer has more than one card art image for BIN, most likely for account range.

For Mastercard (*maximum size: 10*)

```

"cardMetaData": {
  "productID": "9835210843"
}

```

For Visa (*maximum size: 32*):

```

"cardMetaData": {
  "productID": "246380983124"
}

```

Option 3 (for Visa only):

- The Issuer can define the metadata during the digitization by sending the color values and other information.

NOTE: All the parameters highlighted are optional for this Option, the Issuer can send only the **cardArtId** and **termsAndConditionsId**.

```

"CardMetaData": {
  "foregroundColor": "rgb(12,225,585)",
  "backgroundColor": "rgb(13,456,787)",
  "labelColor": "rgb(15,678,679)",
  "shortDescription": "Platinum",
  "longDescription": "Brand X Platinum Elite",
  "contactPhone": "98819838",
  "contactName": "Francisco Pereira",
  "cardArtId": "013",

```

```

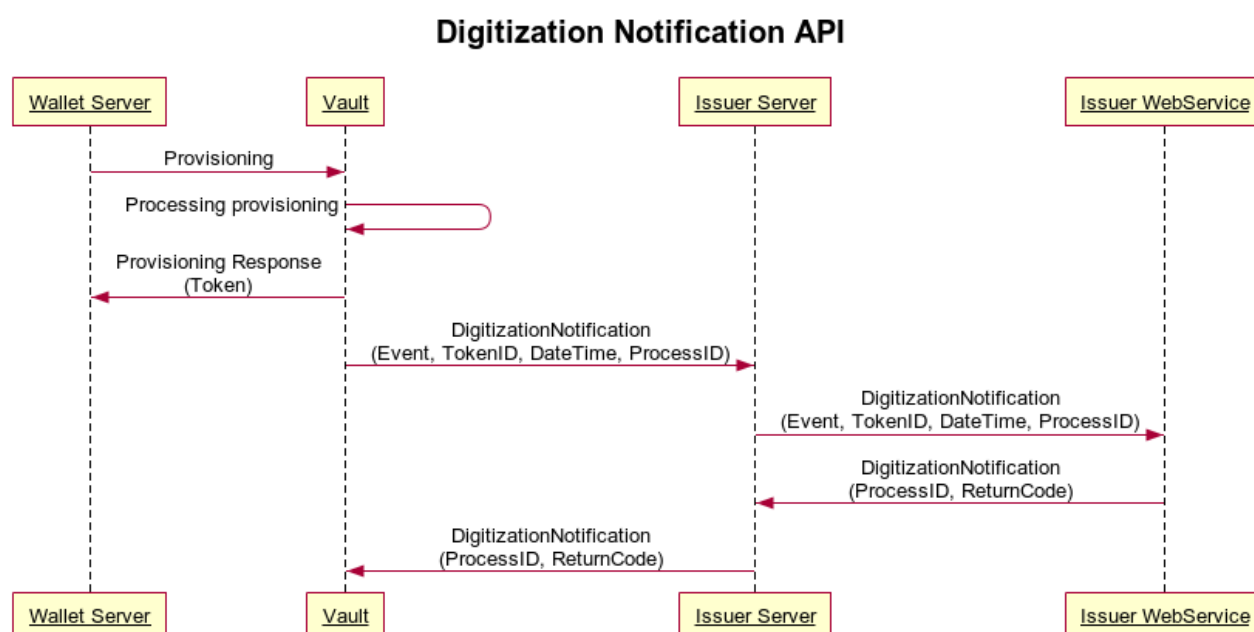
"termsAndConditionsId": "032"
}

```

4.2. DigitizationNotification

This API is used by Issuer server to send notifications to Issuer regarding the digitization process, therefore at the end of the process this API will be triggered **to inform the Issuer** the result of token creation process.

Note: For Mastercard, this is the only API that provides the complete token number associated with the card that was digitized. The calls to GetTokenInfo Inbound API will only retrieve the last 4 digits of the token for this card brand.



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/digitizationnotification	POST
Production: https://{issuer-host:port}/api/v3/digitizationnotification	POST

DigitizationNotificationRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32
Required:	Yes
Element:	processID
Description:	Digitalization process identifier generated for each request. For Mastercard, this field must contain the same value during a complete digitization process, first generated on the CheckEligibility API, and it is sent on the next APIs, such as SendPassCode. For Visa and Amex, this value does not remain the same.
Type:	String
Size:	Max 64
Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; “MDES” – for Mastercard; “AMEX” - for Amex; “PL” – for Private Label. Used to identify the Vault in case tokenRefID does not exist in HST database.
Type:	String
Required:	Yes
Element:	walletID
Description:	Identifier of the wallet that generated the request. Some wallet providers such Apple Pay and Google pay define it with the same value used for device identification. For Apple Pay, this field shows the SEID.
Type:	String
Size:	64
Required:	Optional
Example:	Format at HST White Label Wallet: N3GN-KWH6-NTYC-QNKN
Element:	tokenType
Description:	Possible values are: “HCE”, “SE”, “COF”, “ECOM”, “QRCODE” (Case-Sensitive).
Type:	String
	Required for “VTS” and “MDES”
Required:	Not present for “AMEX”

Element:	dateTime
Description:	Format: yyyy-MM-ddTHH:mm:ss.SSS The value is required to be in GMT.
Type:	String
Required:	Yes
Element:	Event
Description:	Possible values: "CREATED", "STAND_IN" (Case-Sensitive).
Type:	String
Required:	Yes
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Yes
Element:	tokenRequestorName
Description:	Identification of the Token Requestor name requesting digitization. It identifies a Multi Issuer Wallet, an Issuer Wallet or a Merchant, like Uber, Netflix, Adyen, Apple Pay, Samsung Pay and others.
Type:	String
Size:	Max 64
Required:	Optional
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Type:	String
Size:	64
Required:	Yes
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned.

Type:	String
Size:	64
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	encryptedCardInfo
Description:	Encrypted CardInfo related to the card being digitized.
Type:	EncryptedPayload
Required:	Yes
Element:	Source
Description:	How the card number was obtained. Check " <i>CheckEligibility</i> " API for more details. Possible values are: "ON_FILE" – PAN origin is a card number stored in a merchant; "MANUALLY" – PAN was entered by the customer; "MOBILE_APP" – PAN provided by a mobile app. Typically a list of cards provided by the issuer after cardholder authentication; "TOKEN" – The source of pan of this token (ECOM o COF) provisioning was issued by a token device bound (NFC/SE). Applicable to a scenario such as a wallet has an NFC/SE token and it is provisioning a new E-Commerce/COF token. "CONTACTLESS_TAP" - PAN was captured using "Tap to Add Card" service.
Type:	String
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	actionResult
Description:	Result of the digitization process. Possible values are: "APPROVED" – card was successfully tokenized; "APPROVED_IDV" – card was successfully tokenized and will need cardholder authentication for activation; "INVALID_PAN" – the card was not digitized due to the invalid PAN; "INVALID_EXPIRATION_DATE" – the card was not digitized due to the invalid expiration date; "ISSUER_SYSTEM_ERROR" – error on the issuer internal system; "GENERIC_DECLINE" – generic decline on the tokenization process; "ERROR" – error on the tokenization process.
Type:	String

Required:	Yes
Element:	standInReasonCode
Description:	Responsible for informing the Issuer the reason why the digitization was entered in Stand-In flow in case the Issuer system did not respond. Possible values are: “9011” – The line to issuer is down; “9012” – Forced STIP due to no original response from issuer; “9020” – Issuer system time outs; “9027” – CVV2 validate failure following VRM rules defined by the Issuer (ECIP RTD Decline); “9203” – Decline due to Office of Foreign Assets Control (OFAC) embargo; “9206” – Mod-10 check failure; “9208” – Declined because issuing identifier and/or routing identifier is blocked; “9210” – Declined because of issuer participation options; “9212” – Declined due to fraud condition; “9216” – Ineligible data for Token Type. Token is not a device based one; “9217” – Loyalty personalized data input is incorrect; “9601” – Switch detected error.
Type:	String
Required:	Optional for “VTS” and “MDES” Not present for “AMEX”
Element:	termsAndConditions
Description:	Information about the terms and conditions of the card.
Type:	TermsAndConditions
Required:	Optional
Element:	token
Description:	Encrypted TokenInfo of to the token created related to the card being digitized. For the MDES scenario, the Issuer will only receive the token information on this API. EncryptedPayload
Type:	Optional
Required:	
Element:	complementaryAuthenticationInfo
Description:	Additional authentication data. It is used in Apple Multi Device Provisioning scenario, in stand-in flow.
Type:	ComplementaryAuthenticationInfo Object
Required:	Optional

DigitizationNotificationResponse

Element:	requestID
Description:	Request identifier uniquely generated for each request returned by the Issuer.
Type:	String
Required:	Yes
Element:	processID
Description:	Digitalization process identifier generated for each request returned by the Issuer.
Type:	String
Size:	Max 64
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

JSON Examples

DigitizationNotificationRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "vaultIdentification": "VTS",
  "walletID": "N3GN-KWH6-NTYC-QNKN",
  "tokenType": "HCE",
  "dateTime": "2015-05-18T14:40:32.000Z",
  "event": "CREATED",
  "tokenRequestorID": "42301999123",
  "tokenRefID": "DNITHE381502386342002358",
  "PANRefID": "V-3815023863409817870482",
  "encryptedCardInfo": {
    "algorithm": "aes-gcm256",
    "iv": "99aa57b5eb8dc1a8d0f91f40",
    "encryptedData":
"ACBh0D9ZD0k7v1M31uzTk/+7zSNEH9wML7cLi4reKjWcVXm1PFHTz9hxb0RIQdWYBoH7rzyNCHh9l"
```

```

ZA//708BQRgpAIOTY5kgRINWqNiL0DlwKJ+obxGcwssFsBR45ByeiFFFTAk+gPlzM4h4Aj/oqdu4fp
+r0CHiZBTv19PmH4W12BA29lQXI+N",
  "MACLength": 16
},
"source": "MOBILE_APP",
"actionResult": "APPROVED_IDV",
"token": {
  "algorithm": "aes-ccm128",
  "nonce": "b3c0f84e500e50ffcd5f563e",
  "encryptedData":
"Q6sfnucc1f6duTMvzcUa5SueAKUeDpd2Fq+fcSg/xBFU0LhSoiTMJ/3BiZc6uP5GrWbUouoSr01ve
r9YiauDloy9hD4buW2ZiE24sguOpjhlxs2DyNX0ryBlJOjyhK/9z9dfQaRSwK6TxBmndsMAOCGRf5g
QiwifdgF7w/xcJfoDrSnQ9MPkLThyIAA7+y+8ZLiFjjRJGAY1fXjoNnVjsDsxPuIq+p5hI0BrQ9YWH
CqCl1bDX5PycBMT7e5jL2dgz4p7hP2fNr1mXY5EVqhPD12FbjSliXKNib4RdJe/xbol5WCzwhsxcu
+80wt0VMzdZs6DdcrDcMMmB4l+5UAsrzx73JhkAh00j5NK2u+1lrwrAcn8Ul+A/tFv1W3HrarixA1X
PLVpGdOq+3DgjxqkLBZOV1WiZ0D+q0vtVrmkqUvvlyzZafcLufMw9/7KX1s0NmVQDP+2zC1R96VghQ
Njj3wIo7xH/+T0TKhUMqwCapvxkSwD70l87z/eYPKmIb4YXWgbiyKnRUyhCnE5vDxYA10t8+5mz0LY
nJtLAPEMvtyxmIsFU6GW+AYvVJb3ae9ZNfcdsK9DKHpEmHIQ0UffvEA v7ELgjZALW0V1Asx1HiBLJd
YxGX0+3BPuUJssFc1P99AXWyKOTY51KBJMVswxHc=",
  "MACLength": 16
}
}

```

Where:

//Plain CardInfo Object Data:

```

{
  "PAN": "1111110000000003",
  "expirationDate": {
    "month": "11",
    "year": "2024"
  },
  "cardholderName": "FRANCISCO PEREIRA"
}

```

And:

//Plain TokenInfo Object Data:

```

[
  {
    "token": "1111113245678979",
    "expirationDate": {
      "month": "10",
      "year": "2024"
    },
    "state": "ACTIVE",
    "type": "HCE",
    "lastTokenStatusUpdatedTimeStamp": "2015-05-18T14:40:32.000Z",
    "entityOfLastAction": "ISSUER",
    "deviceInfo": {
      "deviceType": "MOBILE_PHONE",
      "deviceNumber": "1234",
      "deviceName": "AndroidCellPhone",
      "serialNumber": "874759678487"
    }
  }
]

```

```

    },
    "OTPCodeIndicate": "PRESENT",
    "OTPCodeExpiration": "2015-05-18T14:40:32.000Z",
    "PANsLastFour": "1234",
    "previousPANsLastFour": "4653",
    "tokenRefID": "DNITHE381502386342002358"
  }
]

```

DigitizationNotificationResponse

```

{
  "requestID": "4",
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "returnCode": "00"
}

```

4.3. SendPassCode

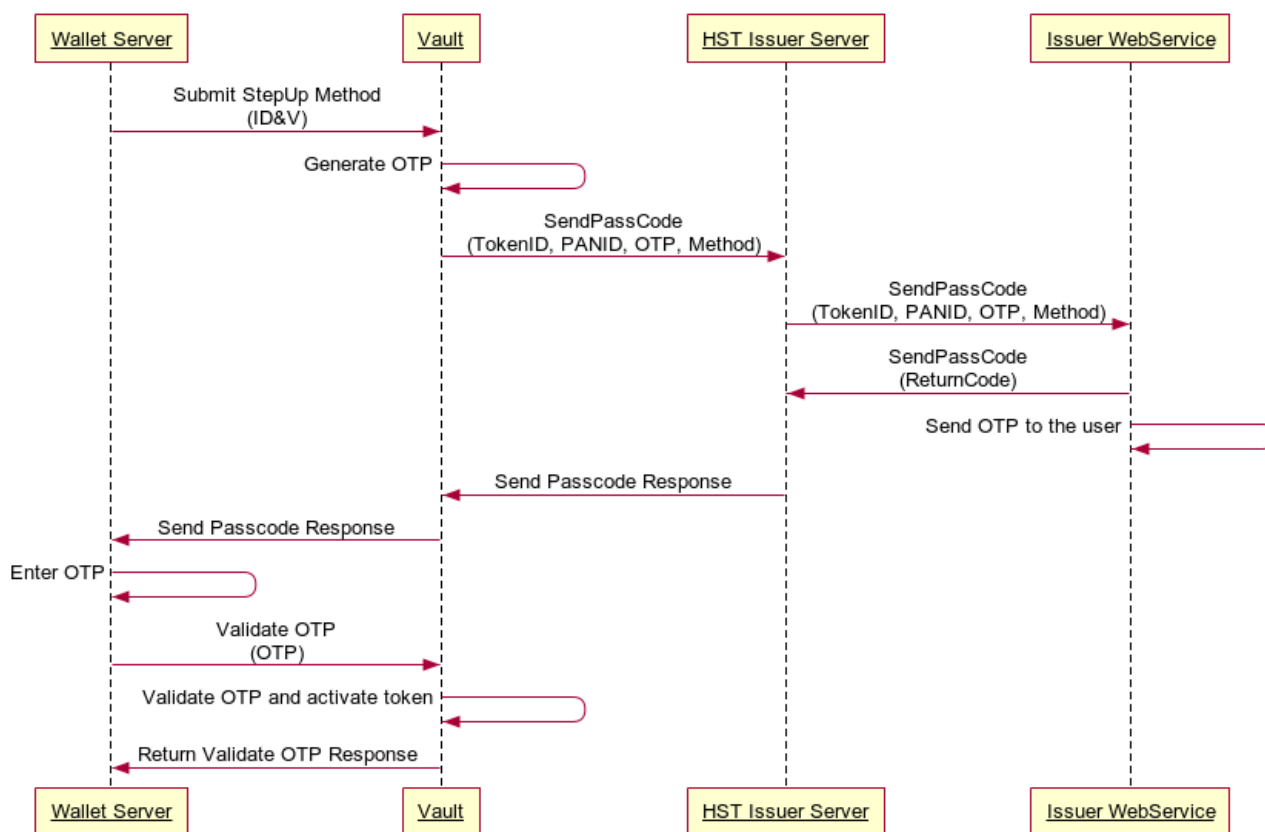
This method is used when Issuer when issuer answers with return code "85" (Requires ID&V) REQUIRE_IDV on CheckEligibility or DeviceBindingEligibility and the cardholder selects "otp_sms" or "otp_email" as step-up methods. In this case the vault generates an OTP and requests the issuer to deliver the OTP to the related phone or email address.

The expected time for response of this API is approximately 2.0 seconds during the requests after being called, otherwise the Vault will receive timeout and the cardholder will get a failed message.

Note: Using the PANRefID or TokenRefID element as a parameter, the Issuer can identify the real card PAN and the respective cardholder that must receive the passcode.

This API is also used to authenticate cardholder post tokenization (Mastercard TAF - Token Authentication Framework), for COF and Click to Pay tokens.

Send Pass Code API



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/sendpasscode	POST
Production: https://{issuer-host:port}/api/v3/sendpasscode	POST

SendPassCodeRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32
Required:	Yes

Element:	processID
Description:	Digitalization process identifier generated for each request. For Mastercard, this field must contain the same value during a complete digitization process, first generated on the CheckEligibility API. For Visa and Amex, this value will not remain the same.
Type:	String
Size:	Max 64
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Type:	String
Size:	64
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated with the Token Requestor, which means it is not unique for a PAN and it can have multiple PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64
Required:	Required for "VTS" Optional for "MDES" Not present for "AMEX"
Element:	authenticationMethod
Description:	Possible values are: "cell_phone" - Text message to account holder's mobile phone number, "email" – E-mail to account holder's email address,

Type:	"push_notification_to_mobile_app" - Push Notification to mobile app instance of cardholder to receive authentication code (OTP). It is only applicable for Post-tokenization authentication (Mastercard TAF).
Required:	String Yes
Element:	OTP
Description:	Authentication code.
Type:	String
Size:	16
Required:	Yes
Element:	OTPEXPIRATION
Description:	Authentication code expiration time. Format: yyyy-MM-dd HH:mm:ss The value will be in GMT.
Type:	String
Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: "VTS" – for Visa; "MDES" – for Mastercard; "AMEX" - for Amex; "PL" – for Private Label. Used to identify the Vault in case tokenRefID does not exist in HST database.
Type:	String
Required:	Yes
Element:	walletID
Description:	Identifier of the wallet that generated the request. Some wallet providers such Apple Pay and Google pay define it with the same value used for device identification. For Apple Pay, this field shows the SEID.
Type:	String
Size:	64
Required:	Optional
Example:	Format at HST White Label Wallet: N3GN-KWH6-NTYC-QNKN
Element:	otpReasonCode
Description:	The possible values are: "CARDHOLDER_STEPUP" - The account holder is being authenticated after adding the card to the token requestor (yellow flow). "PAYMENT" - the account holder is being authenticated to initiate payment transaction. "DEVICE_BINDING" - The account holder is being authenticated to bind the token with bind id as defined in Mastercard TAF.

	“ACCOUNT_BINDING” - The account holder is being authenticated to bind the card number which is tokenized with bind id as defined in Mastercard TAF. “VERIFY_ACCOUNT” - The account holder is being authenticated to verify account ownership. “OTHER” - The account holder is being authenticated for a reason not enumerated in this list.
Type:	String
Size:	Up to 32
Required:	Optional
Element:	encryptedCardInfo
Description:	Encrypted CardInfo . Contains card information to be used on digitization process.
Type:	EncryptedPayload
Required:	Yes
Element:	authenticationRequestID
Description:	The Identifier value to link between multiple authentication requests.
Type:	String
Size:	64
Required:	Conditional – Mastercard only for TAF (Token Authentication Framework). Present for post-tokenization authentication use cases only.

SendPassCodeResponse

Element:	requestID
Description:	Request identifier uniquely generated for each request returned by the Issuer.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: “00” for OK.
Type:	String
Required:	Yes
Element:	processID
Description:	Digitalization process identifier generated for each request returned by the Issuer.
Type:	String
Size:	Max 64

Required:	Yes
-----------	-----

Element:	messageDetail
----------	----------------------

Description:	Detailed response message only for auditing purposes.
--------------	---

Type:	String
-------	--------

Required:	Optional
-----------	----------

Element:	errorDescription
----------	-------------------------

Description:	Error description returned only in error conditions for troubleshooting purposes.
--------------	---

Type:	String
-------	--------

Required:	Optional
-----------	----------

JSON Examples

SendPassCodeRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "tokenRefID": "DNITHE381502386342002358",
  "PANRefID": "V-3815023863409817870482",
  "authenticationMethod": "cell_phone",
  "OTP": "175824",
  "OTPEXpiration": "2015-05-18 14:40:32",
  "vaultIdentification": "VTS",
  "walletID": "N3GN-KWH6-NTYC-QNKN",
  "encryptedCardInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "a96b3e84232d573c6592ceda",
    "encryptedData":
"KV1Mgkv40Nt4yggF1Ka7osdIkyMSsVe8K3o9wpQpMRTGeiXV2I65fIYgjZY1IGEpi/A7+KX3XB8C4
Foo8tEZ5xxQXa2PRudQ9B9s9WZbWoANcyADAw7ix7CQUN4x2ps9+oe8UaLtwjKrbKEDFkCML9rE90
oco7vMr7y+uAlZ2NazPolWwx5fcQkn",
    "MACLength": 16
  }
}
```

Where:

//Plain CardInfo Object Data:

```
{
  "PAN": "1111110000000003",
  "expirationDate": {
    "month": "11",
    "year": "2024"
  },
  "CV2": "500",
}
```

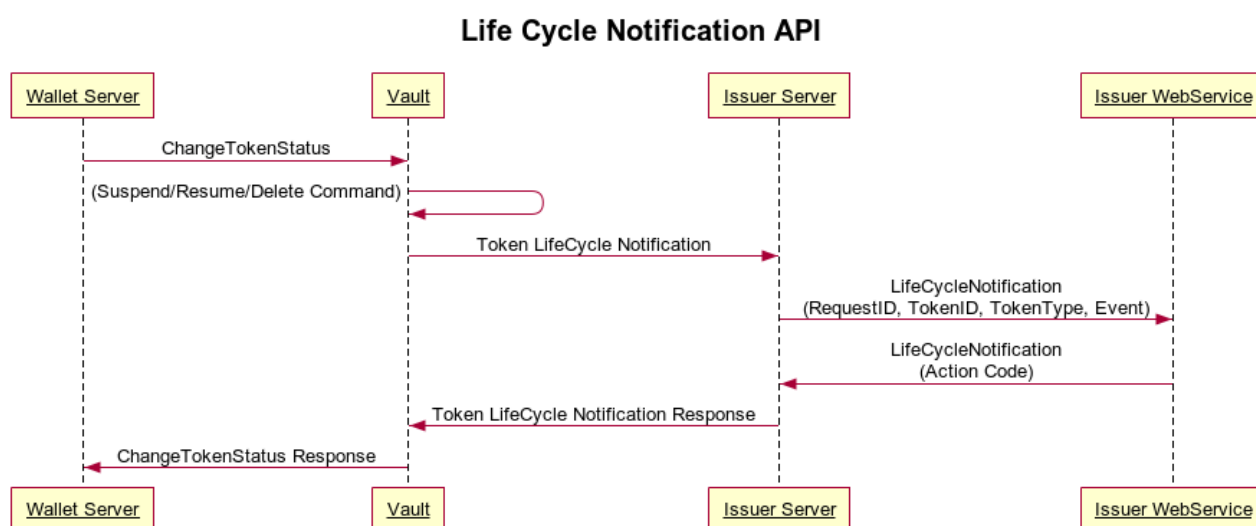
```
{
  "cardholderName": "FRANCISCO PEREIRA"
}
```

SendPassCodeResponse

```
{
  "requestID": "4",
  "returnCode": "00",
  "processID": "1643ef957-622d-4137-abdf-fa605e81e72c",
  "messageDetail": "Passcode received and sent to the user."
}
```

4.4. LifeCycleNotification

This API is used by Issuer server to send some notifications to Issuer to inform it about the life cycle status of tokens. As example, when a token is activated or deactivated this notification will be triggered. It is also used to notify a token bound by the Mastercard TAF service.



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/lifecyclenotification	POST
Production: https://{issuer-host:port}/api/v3/lifecyclenotification	POST

LifeCycleNotificationRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32
Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; “MDES” – for Mastercard; “AMEX” – for Amex; “PL” – for Private Label. Used to identify the Vault in case tokenRefID does not exist in HST database.
Type:	String
Required:	Yes
Element:	walletID
Description:	Identifier of the wallet that generated the request. Some wallet providers such Apple Pay and Google pay define it with the same value used for device identification. For Apple Pay, this field shows the SEID.
Type:	String
Size:	64
Required:	Optional
Example:	Format at HST White Label Wallet: N3GN-KWH6-NTYC-QNKN
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or

	more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Type:	String
Size:	64
Required:	Yes
Element:	tokenType
Description:	Possible values are: "HCE", "SE", "COF", "ECOM", "QRCODE" (Case-Sensitive).
Type:	String
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	dateTime
Description:	Format: yyyy-MM-ddTHH:mm:ss.SSS The value will be in GMT.
Type:	String
Required:	Yes
Element:	event
Description:	Possible values are: "ACTIVATED" – When the token is activated by the vault. "SUSPENDED" – When the token is suspended by the vault. "CANCELLED" – When the token is cancelled by the vault. "INACTIVE" – When the token is inactive, provisioned in yellow flow and now requires further authentication of the cardholder. "DEVICE_BINDING_RESULT" – The token has been attempted to be bound on a trust device (or with bind ID for Mastercard TAF). "ACCOUNT_BINDING" – Account corresponding to the token has been bound with bind ID. "PENDING_ACTIVATION" – Alert triggered to the issuer every 24h notifying the token wasn't activated yet (for Apple). "NO_FIRST_PURCHASE" – reserved for future use, "NO_RECENT_PURCHASE" – reserved for future use, "DELETED_FROM_CONSUMER_APP" – The token has been deleted from the consumer application. The token may still be active (for MDES). "REDIGITIZATION_COMPLETE" – The token has been re-digitized to the device (for MDES). "PAYMENT_ACCOUNT_REFERENCE_UPDATED" – Mastercard only. When tokenizing a Virtual Card Number (VCN), MDES will assign a payment account reference (PAR) value to the VCN. Issuers can opt-in to receive notifications when the PAR value is updated to be consistent across all VCNs linked to the same card account.

Type:	"FUNDING_ACCOUNT_UPDATE" - Token and PAN mapping has been updated due to PAN or expiry or financial account has been changed but token is not changed (for MDES).
Size:	String
Required:	Max 64
Element:	Yes
Description:	tokenUserInfo
Type:	The information of the user that requests the device binding.
Size:	tokenUserInfo Object
Required:	1
	Optional for "VTS" and "MDES"
	Not present for "AMEX"
Element:	merchantInfo
Description:	The information of the merchant that requests the device binding.
Type:	merchantInfo Object
Size:	1
Required:	Optional for "VTS" and "MDES"
	Not present for "AMEX"
Element:	deviceBindingResult
Description:	The possible values are:
	"DEVICE_BINDING_APPROVED" – Approved by green flow.
	"DEVICE_BINDING_OTP" – Approved by yellow flow through OTP method.
	"DEVICE_BINDING_CALL_CENTER" – Approved by yellow flow through Call Center method.
	"DEVICE_BINDING_ISSUER_APP" – Approved by yellow flow through App to App method.
	"DEVICE_BINDING_REMOVED" – The binding between the token and the device was removed.
Type:	String
Required:	Optional for "VTS"
	Optional for "MDES" (TAF use cases)
	Not present for "AMEX".
Element:	deviceInfo
Description:	Information about the device associated with the token.
Type:	DeviceInfo
Required:	Optional
Element:	encryptedCardInfo
Description:	Encrypted CardInfo . Contains card information related to this notification.
Type:	EncryptedPayload
Required:	Optional

Element:	encryptedTokenInfo
Description:	Encrypted TokenInfo . Contains token information related to this notification.
Type:	EncryptedPayload
Required:	Optional
Element:	processID
Description:	Digitalization process identifier generated for each request.
Type:	String
Size:	Max 64
Required:	Optional.
Element:	bindId
Description:	The unique and fixed identifier to a component of data bound to the credential.
Type:	String
Size:	48
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when event value is "DEVICE_BINDING_RESULT" or "ACCOUNT_BINDING".
Element:	certifiedMFAAuthMethodId
Description:	Card brand certified multi-factor authentication method ID that authenticator will use to authenticate cardholder for subsequent authentication after token or Account has been bound with given bind ID.
Type:	String
Size:	1-5
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when event value is "DEVICE_BINDING_RESULT" or "ACCOUNT_BINDING".
Element:	complementaryAuthenticationInfo
Description:	Additional authentication data. It is used in Apple Multi Device Provisioning scenario.
Type:	ComplementaryAuthenticationInfo Object
Required:	Optional

LifeCycleNotificationResponse

Element:	requestID
Description:	Request identifier uniquely generated for each request returned by the Issuer.
Type:	String
Required:	Yes

Element:	returnCode
Description:	Return Code: "00" for OK.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

JSON Examples

LifeCycleNotificationRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "vaultIdentification": "VTS",
  "walletID": "N3GN-KWH6-NTYC-QNKN",
  "tokenRequestorID": "42301999123",
  "tokenRefID": "DNITHE381502386342002358",
  "tokenType": "HCE",
  "dateTime": "2015-05-18T14:40:32.000Z",
  "event": "ACTIVATED"
}
```

LifeCycleNotificationResponse

```
{
  "requestID": "4",
  "returnCode": "00"
}
```

4.5. DeviceBindingEligibility

Device Binding consists of associating a device with an E-Commerce/COF Token. It is an additional layered security with trusted device management. Before initiating the device binding process, it is required by the TR-TSP to complete the enrollment process of that device.

A token can be bound to 100 devices. To distinguish them, the Vault generates a device index in the moment of the device binding.

To complete the device binding process, the Issuer must indicate if the user must or not be verified (green or yellow flow), according to the rules below:

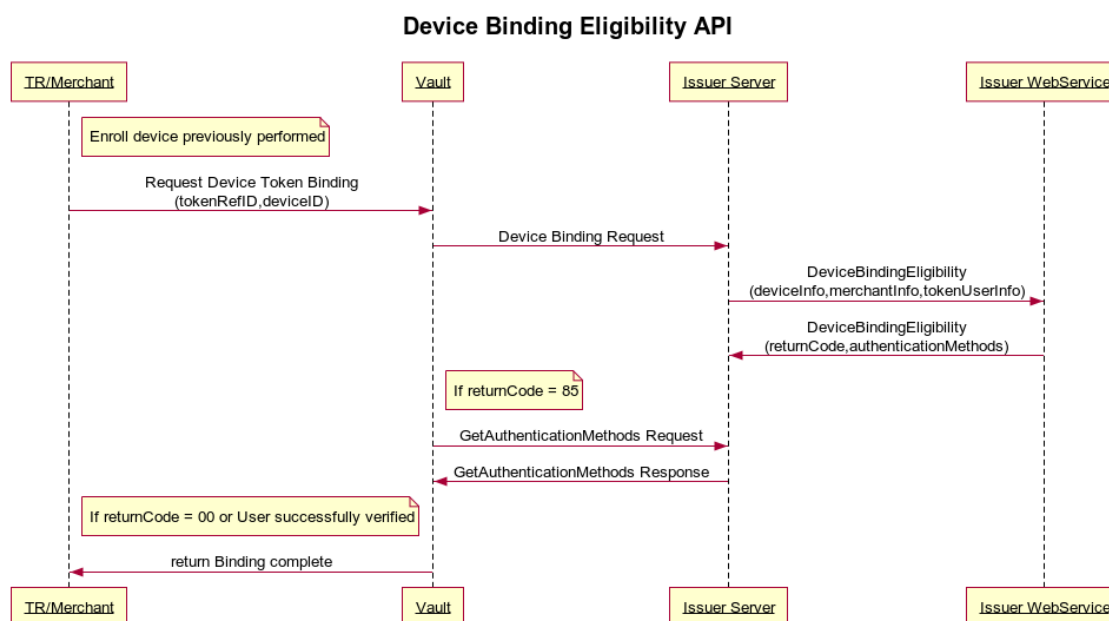
- **Yellow Flow:** The Issuer must return the value **"85"** – **Device is eligible to be bound for this token and cardholder must be verified** in the *returnCode* element to present to the

customer the ID&V methods available for identity validation. Cardholders will receive one or more options to choose from (*Call Center, App to App and OTP*) depending on Issuer implementation.

- **Green flow:** There is no customer ID&V. The Issuer must return the value “00” – **Device is eligible to be bound for this token** in the *returnCode* element.

During the cryptogram request in an E-COM payment flow, if the device is bound, it must be provided in this request the deviceID of the device that is bound to the token. Otherwise, the cryptogram validation will fail.

This API is only used for VISA implementations.



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/devicebindingeligibility	POST
Production: https://{issuer-host:port}/api/v3/devicebindingeligibility	POST

DeviceBindingEligibilityRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Size:	64
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32

Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; “MDES” – for Mastercard; “PL” – for Private Label. Used to identify the Vault in case tokenRefID does not exist in HST database.
Type:	String
Size:	32
Required:	Yes
Element:	walletID
Description:	Identifier of the wallet that generated the request. Some wallet providers such Apple Pay and Google pay define it with the same value used for device identification. For Apple Pay, this field shows the SEID.
Type:	String
Size:	64
Required:	Optional
Example:	Format at HST White Label Wallet: N3GN-KWH6-NTYC-QNKN
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Type:	String
Size:	64

Required:	Yes
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64
Required:	Yes
Element:	deviceInfo
Description:	Data associated with the device. At least, deviceIndex and deviceId will be provided.
Type:	DeviceInfo Object
Required:	Yes
Element:	tokenUserInfo
Description:	The information of the user that requests the device binding.
Type:	TokenUserInfo Object
Required:	Optional
Element:	merchantInfo
Description:	The information of the merchant that requests the device binding.
Type:	MerchantInfo Object
Required:	Optional
Element:	tokenInfo
Description:	Encrypted TokenInfo of to the token created related to the card being digitized.
Type:	EncryptedPayload
Required:	Optional

DeviceBindingEligibilityResponse

Element:	requestID
Description:	Request unique identifier generated for each request returned by the Issuer.
Type:	String
Size:	64
Required:	Yes

Element:	returnCode
Description:	Possible values are: "00" – Device is eligible to be bound for this token (green flow); "05" – Device is not eligible to be bound for this token; "85" – Device is eligible to be bound for this token and cardholder must be verified (yellow flow).
Type:	String
Required:	Yes
Element:	authenticationMethods
Description:	Authentication methods list for specific user's device, if authentication needed. The possible values for implementation are OTP, Call Center and App-to-App and their details are described on the AuthenticationMethod description element.
Type:	Array < AuthenticationMethod >
Required:	Optional, only if returnCode element returns value "85"

JSON Examples

DeviceBindingEligibilityRequest

```
{
  "requestID": "5",
  "institutionCode": "HST",
  "vaultIdentification": "VTS",
  "walletID": "N3GN-KWH6-NTYC-QNKN",
  "tokenRequestorID": "42301999123",
  "tokenRefID": "DNITHE381502386342002358",
  "PANRefID": "V-3815023863409817870482",
  "deviceInfo": {
    "deviceType": "MOBILE_PHONE",
    "deviceNumber": "5355",
    "deviceName": "Mary's Phone",
    "serialNumber": "16344-536536-5453",
    "deviceID": "1234556675587",
    "deviceIndex": "02"
  },
  "tokenUserInfo": {
    "ID": "98765679864",
    "appType": "MOBILE_WEB"
  },
  "merchantInfo": {
    "ID": "12345678",
    "merchantName": "ABC STORE"
  }
}
```

DeviceBindingEligibilityResponse

```
{
  "requestID": "5",
```

```

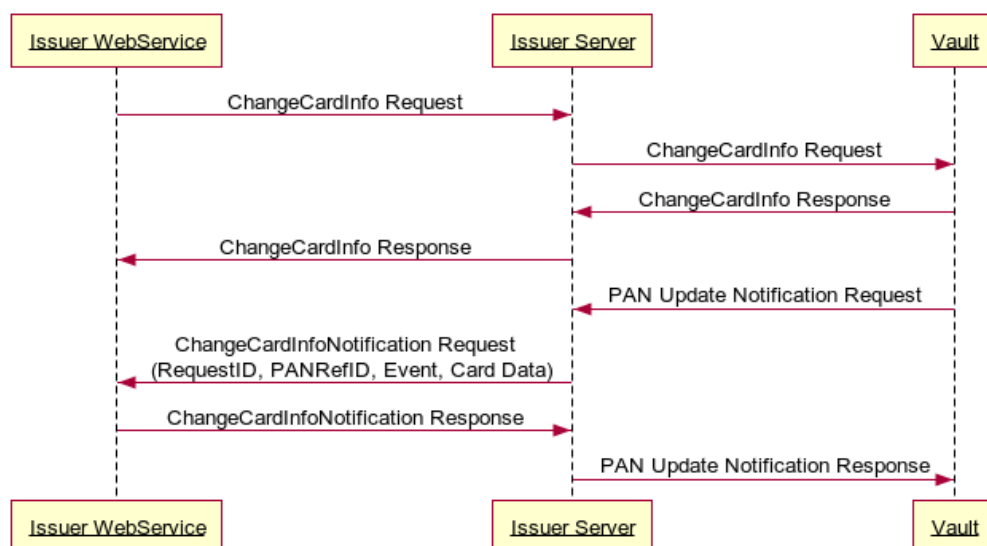
"returnCode": "85",
"authenticationMethods": [
  {
    "identifier": "125485644",
    "type": "bank_app",
    "maskedInfo": "Mobile Banking App",
    "sourceAddress": "com.DemoBank.DemoApp",
    "platform": "ANDROID"
  },
  {
    "identifier": "125485644",
    "type": "cell_phone",
    "maskedInfo": "XXX-XXX-1234"
  }
]
}

```

4.6. ChangeCardInfoNotification

This API is used by Issuer server to send some notifications to Issuer whenever it is performed PAN or PAN expiration date updates. This API is used only in VTS.

Change Card Info Notification API



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/changeCARDinfoNotification	POST

Production: <https://{issuer-host:port}/api/v3/changecardinfoNotification> POST

ChangeCardInfoNotificationRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	32
Required:	Yes
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; Used to identify the Vault. This API is used only by VTS.
Type:	String
Required:	Yes
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated with the Token Requestor, which means it is not unique for a PAN and it can have multiple PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64
Required:	Optional
Element:	dateTime
Description:	Format: yyyy-MM-ddTHH:mm:ss.SSSZ The value will be in GMT.
Type:	String
Required:	Yes
Element:	event
Description:	Possible values are: “PAN_UPDATED” (Case-Sensitive).
Type:	String
Required:	Yes

Element:	messageReasonType
Description:	Possible values are: "ACCOUNT_UPDATE" or "EXP_DATE_UPDATE" (Case-Sensitive).
Type:	String
Required:	Yes
Element:	encryptedOldCardInfo
Description:	CardInfo - Old encrypted card information, containing the current PAN and expiration date.
Type:	EncryptedPayload
Required:	Yes
Element:	encryptedNewCardInfo
Description:	CardInfo - New encrypted card information, containing the new PAN and expiration date.
Type:	EncryptedPayload
Required:	Yes

ChangeCardInfoNotificationResponse

Element:	requestID
Description:	Request unique identifier generated for each request returned by the Issuer.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

JSON Examples

ChangeCardInfoNotificationRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "vaultIdentification": "VTS",
  "PANRefID": "V-3815023863409817870482",
```

```

"dateTime": "2015-05-18T14:40:32.000Z",
"event": "PAN_UPDATED",
"messageReasonType": "EXP_DATE_UPDATED",
"encryptedOldCardInfo": {
  "algorithm": "aes-gcm256",
  "iv": "228be5ada04ab22ae2834fba3f1be459",
  "encryptedData":
"j6RlciEvkUE+LQOusfS0fLdaYt99wnVsfcih9G1190ChD74Zewum6337f+V2WeVcAZjFPm9UZ1B3E
0dpORKFWlFvsYXfjalTv1Y+4X48ie0mIMx5MnLoIg==",
  "MACLength": 16
},
"encryptedNewCardInfo": {
  "algorithm": "aes-gcm256",
  "iv": "e434a9e356425c86338c91bd",
  "encryptedData":
"/rkGCXbH5kib1+hFOa5sMUZV5yckICCs/GT0EkTpFQcJ8xo0/1GBcEQC/vK2UsOBQ/qgILi2I3S0o
NRI5XwnPRg33VjehErWBVjv42nGSUc1Nxyhvg1p0Q==",
  "MACLength": 16
}
}

```

Where:

//Plain OldCardInfo Object Data:

```

{
  "PAN": "1111110000001234",
  "expirationDate": {
    "month": "08",
    "year": "2025"
  }
}

```

//Plain NewCardInfo Object Data:

```

{
  "PAN": "1111110000004321",
  "expirationDate": {
    "month": "05",
    "year": "2026"
  }
}

```

ChangeCardInfoNotificationResponse

```

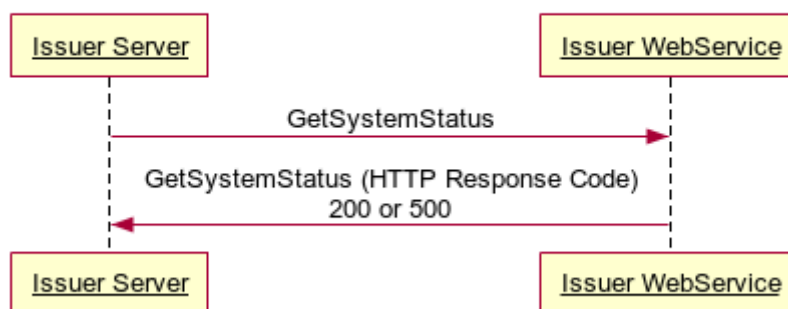
{
  "requestID": "4",
  "returnCode": "00"
}

```


4.7. GetSystemStatus

This API is used by Issuer Server to monitor and to check Issuer system's health status. It is recommended that one of the first APIs to be implemented during an I-TSP/TR-TSP project to establish and validate a connection between Issuer and HST systems.

Get System Status API



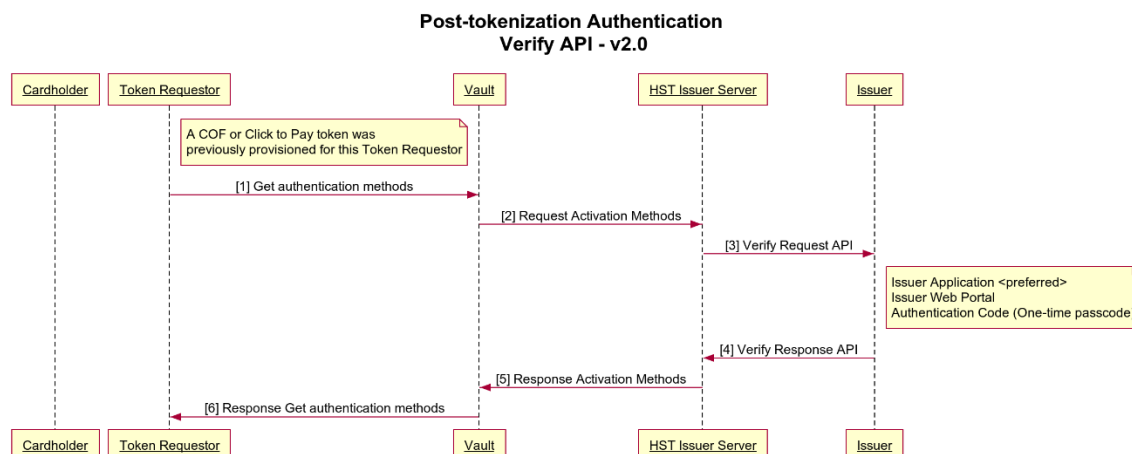
API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/getsystemstatus	GET
Production: https://{issuer-host:port}/api/v3/getsystemstatus	GET

Issuer should respond with 200 if OK or 5XX in case of error or unavailability.

4.8. Verify

This API is invoked when Token Requestor requests post-tokenization authentication.

For now, used for Mastercard TAF (Token Authentication Framework) for COF (Card on file) and Click to Pay tokens, in scenarios such as account holder verification, token binding or before a tokenized payment transaction.



API endpoint	Method
Sandbox: https://{sandbox-issuer-host:port}/api/v3/verify	POST
Production: https://{issuer-host:port}/api/v3/verify	POST

VerifyRequest

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Size:	Max 64
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	Max 32
Required:	Yes
Element:	apiVersion
Description:	Indicate the version of the API that must be used.
Type:	String
Size:	Max 16
Required:	Optional
Element:	serviceld
Description:	Unique identifier to choose the service needed. Options are: 1 – TK 2 – ACS
Type:	String
Size:	Max 2
Required:	Yes
Element:	Brand
Description:	Possible values are: 1 – Visa 2 – Mastercard 3 – Amex 4 – Elo 5 – Carnet
Type:	String
Size:	Max 2
Required:	Yes

Element:	processID
Description:	Digitalization process identifier generated for each request.
Type:	String
Size:	Max 64
Required:	Yes
Element:	paymentData
Description:	Contains payment meta information.
Type:	PaymentData Object.
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when reasonCode value is "PAYMENT".
Element:	deviceInfo
Description:	Data is associated with the device.
Type:	DeviceInfo Object
Required:	Optional
Element:	tokenInfo
Description:	Encrypted TokenInfo of to the token created related to the card being digitized.
Type:	EncryptedPayload
Required:	Optional
Element:	riskDecisioningInfo
Description:	Contains information about the decision recommended by the Token Requestor.
Type:	RiskInformation Object.
Required:	Conditional
Element:	tkData
Description:	Contains tokenization flow information.
Type:	TKData Object
Required:	Conditional

VerifyResponse

Element:	requestID
Description:	Request identifier uniquely generated for each request returned by the Issuer.
Type:	String
Size:	Max 64
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Size:	Max 256
Required:	Optional

Element:	returnCode
Description:	Possible values are: "0" – Requested action approved, and cardholder authentication is not required; "1" – Cardholder authentication is required; "3" – Requested action is not approved; "4" – Token not found; "5" – Token cancelled; "6" – Token blocked; "7" – Token expired; "98" – Invalid request; "99" – Internal error.
Type:	String
Size:	Max 3
Required:	Yes
Element:	challengeMethod
Description:	Challenge methods data for specific transaction.
Type:	Array < AuthenticationMethod >
Required:	Optional, only if returnCode element returns value "1".

5. Tokenization BUS – Inbound (From Issuer to HST)

The HST Tokenization BUS webservice is designed to allow issuers to integrate its current CMS (Card Management System) or Internet Banking directly with the Issuer server. In such way, it is possible to perform a series of operations within its own platform.

5.1. GetAssociatedTokens

This API is used to get the Token Reference IDs associated with a PAN, PAN Reference ID and/or UserID. Then, it is necessary to call [GetTokenInfo](#) to obtain details about the token. For Issuer Wallets it's also possible to search for tokens associated with an UserID previously defined on GetAvailableCards or AuthenticateCardholder (Issuer Wallet APIs described in other documentation).

- 1-) In case PAN and PAN Reference ID elements were both sent during request, only the PAN Reference ID will be used, and PAN element will be ignored.
- 2-) If the request is performed using PAN element, the results can return **all** the tokensReferenceIDs associated to the cardholder, regardless of the device.
- 3-) If the request is performed using PANRefID element, only the tokens associated to such PANRefID will be returned. This PANRefID will be related to a unique device, i.e., it cannot be related to all the tokens associated to the cardholder.
- 4-) Since the UserID is an information not available in the Vault, if this element is used in the search, the inquiry will be performed **only** in the HST Environment local database, for associated tokens retrieving. In this case the Issuer Server will not send the request to the Vault(s).
- 5-) If multiple elements are provided in the request message, the priority order used during the search will be (from the highest to lowest): userID, PANRefID and cyphered PAN elements, respectively.

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/getassociatedtokens	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/getassociatedtokens	POST

GetAssociatedTokensRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes

Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes
Element:	tokenRequestorID
Description:	The token requestor associated with the token. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet, an Issuer Wallet or a Merchant. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project. If provided, results will only contain tokens related to that specific Token Requestor ID.
Type:	String
Required:	Optional
Element:	tokenType
Description:	Results will only contain tokens of the specified type. Possible values are: "HCE", "SE", "COF", "ECOM", "QRCODE" (Case-Sensitive).
Type:	String
Size:	32
Required:	Optional
Element:	userID
Description:	Issuer identification of the cardholder. Typically, an account or online banking user ID defined in response of GetAvailableCards or AuthenticateCardholder. Only for auditing purposes on HST's system, there is no participation during the provisioning and transaction flows.
Type:	String
Required:	Optional
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated with the Token Requestor, which means it is not unique for a PAN and it can have multiple PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64

Required:	Optional
Element:	encryptedCardInfo
Description:	Contains a PAN number on an encrypted CardInfo object. PAN is the only attribute of CardInfo that must be populated.
Type:	EncryptedPayload
Required:	Optional for “VTS” and “MDES” Required for “AMEX”
Element:	tokenState
Description:	Searches for tokens in a specific state. Possible values are: “ACTIVE”, “SUSPENDED”, “INACTIVE”, “CANCELED” (Case-Sensitive).
Type:	String
Size:	0-32
Required:	Optional for “VTS” and “MDES” Not present for “AMEX”
Element:	operatorID
Description:	The operator identification code.
Type:	String
Size:	0-16
Required:	Required for VTS and MDES. Not present otherwise.
Element:	operatorName
Description:	Operator name.
Type:	String
Size:	0-200
Required:	Required for MDES. Not present otherwise.
Element:	operatorPhone
Description:	Operator’s contact phone.
Type:	String
Size:	0-20
Required:	Required for MDES. Not present otherwise.
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; “MDES” – for Mastercard; “AMEX” – for Amex; “PL” – for Private Label. Used to identify the Vault in case the tokenRefID does not exist in HST database.
Type:	String
Size:	32
Required:	Required for “AMEX”
Element:	cardKey

Description:	ID of the internal Amex card, in case the issuer has this data
Type:	String
Size	64
Required:	Optional for "AMEX" Not present for "VTS" and "MDES"

GetAssociatedTokensResponse

Element:	requestID
Description:	Request unique identifier generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK. Return Code: "01" for Ok with a warning condition – Check Error description for more information.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional
Element:	tokenRefIDList
Description:	List of Token Reference IDs.
Type:	Array <String>
Required:	Optional
Element:	tokenInfoList
Description:	List of encrypted TokenInfo objects. This list has paired indexes with tokenRefIDList elements.
Type:	Array < EncryptedPayload >
Required:	Optional

JSON Examples

GetAssociatedTokensRequest

```
{
  "requestID": "9",
  "institutionCode": "HST",
  "tokenRequestorID": "42301999123",
  "tokenType": "HCE",
  "PANRefID": "V-3815023863409817870482",
  "encryptedCardInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "a96b3e84232d573c6592ceda",
```



```

    "encryptedData":
    "KV1Mgkv40Nt4yggF1Ka7osdIkyMSsVe8K3o9wpQpMRTGeiXV2I65fIYgjZY1IGEpj/A7+KX3XB8C4
    Foo8tEZ5xxQXa2PRudQ9B9s9WZbWoANcyaDAdw7ix7CQUN4x2ps9+oe8UaLtwjKrbKEDFkCML9rE90
    oco7vMr7y+uAlZ2NazPoWwx5fcQkn",
    "MACLength": 16
  },
  "tokenState": "INACTIVE"
}

```

Where:

//Plain CardInfo Object Data:

```

{
  "PAN": "1111110000000003",
  "expirationDate": {
    "month": "11",
    "year": "2024"
  },
  "CVV2": "500",
  "cardholderName": "FRANCISCO PEREIRA"
}

```

GetAssociatedTokensResponse

```

{
  "requestID": "9",
  "returnCode": "00",
  "tokenRefIDList": [
    "DNITHE381502386342002358",
    "A4N6HKA45114456AS4584844"
  ],
  "tokenInfoList": [
    {
      "algorithm": "aes-gcm256",
      "iv": "F6721F7B3A63A8F4908CF5245B154120",
      "encryptedData": "*****...",
      "MACLength": 12
    },
    {
      "algorithm": "aes-gcm256",
      "iv": "ECAE3F12E0E73177A030084B265EE055",
      "encryptedData": "*****...",
      "MACLength": 12
    }
  ]
}

```

Where:

//Plain TokenInfo Object Data:

```

{
  "token": "1111113245678979",
  "expirationDate": {
    "month": "10",

```

```

    "year": "2024"
  },
  "state": "ACTIVE",
  "type": "HCE",
  "lastTokenStatusUpdatedTimeStamp": "2015-05-18T14:40:32.000Z",
  "entityOfLastAction": "ISSUER",
  "deviceInfo": {
    "deviceType": "MOBILE_PHONE",
    "deviceNumber": "1234",
    "deviceName": "AndroidCellPhone",
    "serialNumber": "874759678487"
  },
  "OTPCodeIndicate": "PRESENT",
  "OTPCodeExpiration": "2015-05-18T14:40:32.000Z",
  "PANsLastFour": "1234",
  "previousPANsLastFour": "4653",
  "tokenRequestorID": "42301999123",
  "tokenRefID": "DNITHE381502386342002358"
}

```

5.2. GetTokenInfo

This API is used to get all the information about a token. During request, the tokenRefID must be sent.

For Mastercard, the token value will not be provided because the brand doesn't provide this value in this API. The only API the issuer may receive the token is DigitizationNotification.

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/gettokeninfo	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/gettokeninfo	POST

GetTokenInfoRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes

Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Type:	String
Size:	64
Required:	Required for "MDES", "VTS" and "AMEX"
Element:	deviceBindingInfo
Description:	True if it must return device binding data or false if not. By default, it is False. If True, deviceInfo Object list must return on the response deviceIDs and deviceIndexes bound to the Token.
Type:	Boolean
Required:	Used only for "VTS" Not present for "MDES" and "AMEX"
Element:	vaultIdentification
Description:	Possible values are: "VTS" – for Visa; "MDES" – for Mastercard; "AMEX" – for Amex; "PL" – for Private Label. Used to identify the Vault in case the tokenRefID does not exist in HST database.
Type:	String
Size:	32
Required:	Required for "AMEX" Optional for "VTS" and "MDES"
Element:	operatorID
Description:	The operator identification code.
Type:	String
Size:	0-16
Required:	Required for VTS and MDES. Not present otherwise.
Element:	operatorName
Description:	Operator name.
Type:	String
Size:	0-200
Required:	Required for MDES. Not present otherwise.

Element:	operatorPhone
Description:	Operators contact phone.
Type:	String
Size:	0-20
Required:	Required for MDES. Not present otherwise.
Element:	tokenRequestorID
Description:	The token requestor associated with the token. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet, an Issuer Wallet or a Merchant. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Required for VTS, Optional for MDES

GetTokenInfoResponse

Element:	requestID
Description:	Request unique identifier generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK "92" for Token Not Found "95" for Cryptography Error "96" for Invalid Data "97" for Required Data Missing "98" for Invalid Request "99" for System Error, please check error description.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional
Element:	tokenInfo
Description:	Encrypted list of TokenInfo objects related to the requested tokenRefIDs.
Type:	EncryptedPayload
Required:	Optional

Element:	deviceInfo
Description:	List of DeviceInfo Objects related to the requested tokenRefID. It is returned if deviceBindingInfo element in the request is True. The list of DeviceInfo objects will contain only the deviceId and deviceIndex elements.
Type:	List of deviceInfo Objects
Required:	Optional
Element:	tokenRequestorID
Description:	Identification of the Token Requestor associated with the token.
Type:	String
Required:	Optional
Element:	RiskInformation
Description:	RiskData provided by the Token Requestor on digitization process.
Type:	RiskInformation Object
Required:	Optional

JSON Examples

GetTokenInfoRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "tokenRefID": "DNITHE381502386342002358",
  "deviceBindingInfo": "True"
}
```

GetTokenInfoResponse

```
{
  "requestID": "4",
  "returnCode": "00",
  "tokenInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "b3c0f84e500e50ffcd5f563e",
    "encryptedData":
"Q6sfnucc1f6duTMvzcUa5SueAKUeDpd2Fq+fcSg/xBFU0LhSoiTMJ/3BiZc6uP5GrWbUouoSr01ve
r9YiauDloy9hD4buW2ZiE24sguOpjhlx2DyNX0ryB1JOjyhK/9z9dfQaRSwK6TxBmndsMAOCGRf5g
QiwifdgF7w/xcJfoDrSnQ9MPkLThyIAA7+y+8ZLiFjjRJGAY1fXjoNnVjsDsxpuiq+p5hI0BrQ9YWH
CqC1lbDX5PycBMT7e5jL2dgz4p7hP2fNr1mXY5EVqhPD12FbjSLiXKNib4RdJe/xbol5WCzwhsxcu
+80wt0VMzdZs6DdcrDcMMmB4l+5UAsrzz73JhkAh00j5NK2u+llrwrAcn8U1+A/tFv1W3HrarixA1X
PLVpGdOq+3DgjxqkLBZOV1WiZ0D+q0vtVrmkqUvvlyzZafcLufMw9/7KX1s0NmvQDP+2zC1R96VghQ
Njj3wIo7xH/+T0TKhUMqwCapvxkSwD70187z/eYpKmIb4YXWgbiyKnRUyhCnE5vDxYA10t8+5mz0LY
nJtLAPeMvtyxmIsFU6GW+AYvVJb3ae9ZNfcdsK9DkHpEmHIQ0UffvEAv7ELgjZALWOV1Asx1HiBLJd
YxGX0+3BPuUJssFc1P99AXWyKOTY51KBJMVswXhc=",
    "MACLength": 16
  },
  "tokenRequestorID": "42301999123",
  "deviceInfo": [
    {
```

```

    "deviceID": "87755776656",
    "deviceIndex": "01"
  }
]
}

```

Where:

//Plain TokenInfo Object Data:

```

[
  {
    "token": "1111113245678979",
    "expirationDate": {
      "month": "10",
      "year": "2024"
    },
    "state": "ACTIVE",
    "type": "HCE",
    "lastTokenStatusUpdatedTimeStamp": "2015-05-18T14:40:32.000Z",
    "entityOfLastAction": "ISSUER",
    "deviceInfo": {
      "deviceType": "MOBILE_PHONE",
      "deviceNumber": "1234",
      "deviceName": "AndroidCellPhone",
      "serialNumber": "874759678487"
    },
    "OTPCodeIndicate": "PRESENT",
    "OTPCodeExpiration": "2015-05-18T14:40:32.000Z",
    "PANsLastFour": "1234",
    "previousPANsLastFour": "4653",
    "tokenRefID": "DNITHE381502386342002358"
  }
]

```

5.3. ChangeTokenStatus

This API is used to change the status of a token. Issuer must inform the Token Reference ID to perform the operation. Through this API, it is possible to activate, suspend, resume and delete a token. The conditions are described below:

- A token can be **activated** from inactive status after a cardholder verification is performed by the Issuer.
- A token may be **suspended** because of a stolen/lost device or card. Once is submitted with a suspension reason, the status is changed to “suspended” and the token can no longer be used for payments unless it is activated again.
- A token can be **reactivated (resumed)** from a suspension after the cardholder recovers a lost device/card and requests the activation to the Issuer.
- Any token can be **deleted** due to cardholder reasons (*lost/stolen card or device, closed PAN, etc.*) regardless of the actual token status. Once a token is deleted, it can no longer be used for payments or activated again (*).



Besides token status lifecycle, this API can also be used to manage the **Device binding** lifecycle, allowing token **device binding approval and removal operations**.

(*) Please be aware that VTS will provision the same tokenRefID for a HCE token when occurs within seven days from the token deletion by the cardholder. It doesn't mean it is possible to revert a deleted token from lifecycle operations.

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/changetokenstatus	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/changetokenstatus	POST

ChangeTokenStatusRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for

Type:	further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others.
Size:	Contains the Token Reference ID that is subject to the status change.
Required:	String
	64
	Yes
Element:	action
Description:	Possible values are: "DELETE", "SUSPEND", "RESUME", "DEVICE_BINDING_APPROVE", "DEVICE_BINDING_REMOVE".
Type:	String
Required:	Yes
Element:	operatorID
Description:	The operator identification code.
Type:	String
Size:	0-16
Required:	Required for VTS and MDES. Not present otherwise.
Element:	operatorName
Description:	Operator name.
Type:	String
Size:	0-200
Required:	Required for MDES. Not present otherwise.
Element:	operatorPhone
Description:	Operators contact phone.
Type:	String
Size:	0-20
Required:	Required for MDES. Not present otherwise.
Element:	reason
Description:	The reason why the change was made.
Type:	String
Size:	256
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	additionalInformation
Description:	During the change token status process, it is possible to add more information if desired by the helpdesk operator to complement the reason already indicated.
Type:	String
Size:	0-256
Required:	Optional
Element:	vaultIdentification
Description:	Possible values are: "VTS" – for Visa; "MDES" – for Mastercard; "AMEX" – for Amex;

Type:	String
Size:	32
Required:	Optional for "VTS" and "MDES" Required for "AMEX"
Element:	deviceInfo
Description:	Only valid for <i>Device Binding</i> lifecycle operations. Only deviceID and deviceIndex must be informed.
Type:	DeviceInfo Object
Required:	Optional para "VTS" Not present for "MDES" and "AMEX"
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project.
Type:	String
Size:	64
Required:	Required for "VTS". Optional for "MDES" Not present for "AMEX"

ChangeTokenStatusResponse

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK. "91" for Invalid Token Status/Token Not Active "92" for Token Not Found "93" for Token Already in the State Requested "95" for Cryptography Error "96" for Invalid Data "97" for Required Data Missing "98" for Invalid Request "99" for System Error, please check error description
Type:	String
Required:	Yes

Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

JSON Examples

ChangeTokenStatusRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "tokenRefID": "DNITHE381502386342002358",
  "action": "DELETE",
  "operatorID": "134",
  "reason": "About to expire"
}
```

ChangeTokenStatusResponse

```
{
  "requestID": "4",
  "returnCode": "00"
}
```

5.4. ActivateToken

This API is used to activate a token informing the Token Reference ID during ID&V flow (call center inbound call, call center outbound call or App to App).

Mostly used to activate a token on a digitalization flow that requires cardholder identification and verification (ID&V) with the authentication method App-to-App (if used by the Issuer) during the yellow flow. Also can be invoked by Issuer Card Management tools (Helpdesk).

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/activatetoken	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/activatetoken	POST

ActivateTokenRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) on the specified device (WalletID/DeviceID). It is a value assigned by the vault. Each token generates a tokenRefID value, which means a PAN can have one or more tokenRefID values assigned. By using this data, it is not necessary to input the real token value. It is recommended to associate or to bind the tokenRefID value with the PANRefID for further use in APIs such as Get TokenInfo, ChangeTokenStatus, ActivateToken and others. Token Reference ID associated with the token being activated. This element must be sent when the cardholder decides to activate a single token.
Type:	String
Size:	64
Required:	Yes
Element:	activationCode
Description:	This can be a random code generated by the issuer only as an auditing purpose to be associated to the successful activation process and it is not validated by the Vault.
Type:	String
Size:	0-16
Required:	Deprecated
Element:	operatorID
Description:	The operator identification code.
Type:	String
Size:	0-16
Required:	Required for VTS and MDES. Not present otherwise.
Element:	operatorName
Description:	Operator name.
Type:	String
Size:	0-200
Required:	Required for MDES. Not present otherwise.

Element:	operatorPhone
Description:	Operators contact phone.
Type:	String
Size:	0-20
Required:	Required for MDES. Not present otherwise.
Element:	reason
Description:	The reason why the activation was made.
Type:	String
Size:	Up to 254 characters
Required:	Required for Visa and Mastercard, not present for Amex.
Element:	vaultIdentification
Description:	Possible values are: “VTS” – for Visa; “MDES” – for Mastercard; “AMEX” – for Amex; “PL” – for Private Label. Used to identify the Vault in case the tokenRefID does not exist in HST database.
Type:	String
Size:	32
Required:	Optional for “VTS” and “MDES” Required for “AMEX”
Element:	tokenRequestorID
Description:	The token requestor associated with the token. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet, an Issuer Wallet or a Merchant.
Type:	String
Size:	64
Required:	Required

ActivateTokenResponse

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: “00” for OK.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

JSON Examples

ActivateTokenRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "tokenRefID": "DNITHE381502386342002358",
  "operatorID": "14",
  "reason": "Token activation pending"
}
```

ActivateTokenResponse

```
{
  "requestID": "4",
  "returnCode": "00"
}
```

5.5. GetPANByPANRefID

This API is a helper function exclusive for the App-to-App authentication. Issuers using this step-up method need to retrieve PAN information based on PAN Reference ID to verify if the card being digitized is related to the cardholder being authenticated on the Issuer app. During App to App step-up method the Issuer receives a PANRefID value on its mobile application, and through this API the Issuer can get the PAN to validate if the cardholder has the PAN that is trying to digitize.

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/getpanbypanrefid	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/getpanbypanrefid	POST

GetPANByPANRefIDRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes

Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated with the Token Requestor, which means it is not unique for a PAN and it can have multiple PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Required:	Required for "VTS" and "MDES" Not present for "AMEX"

GetPANByPANRefIDResponse

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Code: "00" for OK.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional
Element:	encryptedPAN
Description:	Encrypted PAN Number. Contains a string containing the card PAN related to the PAN Reference ID.
Type:	EncryptedPayload
Required:	Yes

JSON Examples

GetPANByPANRefIDRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "PANRefID": "V-3815023863409817870482"
}
```

GetPANByPANRefIDResponse

```
{
  "requestID": "4",
  "returnCode": "00",
  "encryptedPAN": {
    "algorithm": "aes-gcm256",
    "iv": "515B6D4BC91BDA4E8FFF1D5D246657AB",
    "encryptedData":
    "8ZqX1V9oDgJfOUqHdam7nwtWgT595qDN+T1QFIGc4/Jzw6McJKW2FWsr",
    "MACLength": 16
  }
}
```

Where:

//Plain CardInfo Object Data:

```
{
  "PAN": "1111110000000003"
}
```

5.6. ChangeCardInfo

This API is used either to replace an old PAN for a new PAN in such a way that all existing tokens will be tied with the new PAN and the cardholder doesn't need to provision again.

After the process is executed, the Issuer will receive the new PAN when the user performs a transaction with the existing tokens.

Moreover, the Issuer can also extend the expiration date for a current card.

These are the use conditions for this API:

- To replace a PAN:
 - In the request message the PANs and expiration dates must be provided in both objects *encryptedOldCardInfo* and *encryptedNewCardInfo*.
 - **The new PAN must not have any associated tokens, i.e., the new card must not have been digitized yet in any other wallet or merchant.**
 - **Based on the previous condition, it's highly recommended the execution of this command before providing the new card to the cardholder or before the cardholder activates it.**

➤ For Visa Cards, when replacing an existing PAN with a new one, if there are tokens associated with the existing PAN, the new PAN must have the same BIN as the original PAN. If the existing PAN and the replacement PAN have different BINs, you must delete the tokens on the existing PAN before calling the ChangeCardInfo API to replace the PAN.

- To extend the expiration date:

➤ In the request message the old card expiration date must be provided in *encryptedOldCardInfo* object, and the new card expiration date must be filled in *encryptedNewCardInfo* object. The same PAN must be given in both objects.

API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/changecardinfo	POST
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/changecardinfo	POST

ChangeCardInfoRequest

Element:	requestID
Description:	Request unique identifier generated for each request by the Issuer.
Type:	String
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Required:	Yes
Element:	operation
Description:	Possible values are "UPDATE", "UNLOCK", "SUSPEND", "RESUME" or "DELETE".
	Notes:
	• "UPDATE" operation is used by VAU/VTs, MDES and AMEX. • "UNLOCK", "SUSPEND" and "RESUME" operations are used only for AMEX. • "DELETE" only used for PAN Update (Reversal) operation from VAU/VTs.
	See notes (*) for usage details.
Type:	String
Required:	Yes
Element:	operatorID
Description:	The operator identification code.
Type:	String
Size:	0-16

Required:	Required for VTS and MDES. Not present otherwise.
Element:	operatorName
Description:	Operator name.
Type:	String
Size:	0-200
Required:	Required for MDES. Not present otherwise.
Element:	operatorPhone
Description:	Operators contact phone.
Type:	String
Size:	0-20
Required:	Required for MDES. Not present otherwise.
Element:	reason
Description:	The reason why a change was made.
Type:	String
Required:	Required for “VTS” and “MDES” Not present for “AMEX”
Element:	encryptedOldCardInfo
Description:	CardInfo - Old encrypted card information. See notes (*) for usage details.
Type:	EncryptedPayload
Required:	Yes
Element:	encryptedNewCardInfo
Description:	CardInfo - New encrypted card information. See notes (*) for usage details.
Type:	EncryptedPayload
Required:	Conditional
Element:	operationReasonCode
Description:	A reason code for the operation chosen by the issuer. One of the possible values are valid: • “ACCOUNT_CLOSED” • “CONTACT_CARDHOLDER” • “PORTFOLIO_CONVERSION” • “VAU_CARDHOLDER_OPT_IN” • “VAU_CARDHOLDER_OPT_OUT”
Type:	String
Required:	Optional – Only used for VAU operations.

ChangeCardInfoResponse

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Required:	Yes
Element:	returnCode
Description:	Return Codes: "00" for Ok. "94" for Invalid Replacement PAN "95" for Cryptography Error "96" for Invalid Data "97" for Required Data Missing "99" for System Error, please check error description.
Type:	String
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Required:	Optional

(*) Attention required:

To update card information, it is mandatory to inform in the objects **encryptedOldCardInfo** and **encryptedNewCardInfo** the elements accordingly to the expected scenario, as indicated below:

1-) To update PAN and ExpirationDate is required to inform: Old PAN, Old Expiration Date, New PAN, New Expiration Date.

2-) To update only ExpirationDate is required to inform: Old PAN, Old Expiration Date, Old PAN, New Expiration Date.

3-) Cardholder required info for VAU operations:

Lifecycle Operation	Operation Type	Old PAN	Old Exp Date	New PAN	New Exp Date
<i>PAN Update with Exp Date</i>	UPDATE	Required	Required	Required	Required
<i>PAN Update (Reversal)</i>	DELETE	Required	-	-	-
<i>Exp Date Update</i>	UPDATE	Required	Required	Old PAN*	Required
<i>Close Account</i>	UPDATE	Required	-	-	-
<i>Contact Cardholder</i>	UPDATE	Required	-	-	-
<i>Opt-In VAU</i>	UPDATE	Required	-	-	-
<i>Opt-Out VAU</i>	UPDATE	Required	-	-	-

JSON Examples

ChangeCardInfoRequest

```
{
  "requestID": "4",
  "institutionCode": "HST",
  "operation": "UPDATE",
  "reason": "About to expire",
  "operatorID": "12",
  "encryptedOldCardInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "508ad7193d0b634647cdd931",
    "encryptedData":
"8ztAmsfoQdE7P22LqdAJD24VdoQay5k6mdghbKRQsPNqcNnjyl+MqDTvqqQITgolhtMawvDjnn3f0
m0JfJDvW8EeTs5ZcutGs68IKMlRGf0+xrQBf08iXAKKEDs0qksyuj0Jm3bvWpAyXmSe4NIki40c+T8
plK8g/KPFHElDZVq6gJ329zmWhOMkc6GnN/Kz",
    "MACLength": 16
  },
  "encryptedNewCardInfo": {
    "algorithm": "aes-ccm128",
    "nonce": "e434a9e356425c86338c91bd",
    "encryptedData":
"H0njeQMSPId0iuS0sILBind0GkUetIg4BoY1U+rXwf4yxeXr5f0wTru531l6acVhZvXwqWP4xqDRG
qfQ88LN52dmt+ZfiuA2KbcPszjWkRrImg0q/tFJAuhw1KdkCcwS8+vNLrLv56H32PB8vfJizkL0zf
/e5Y2X5jNyp7FF/D4+UHZMZzfBUA8HyQDcZ9g",
    "MACLength": 16
  }
}
```

Where:

//Plain OldCardInfo Object Data:

```
{
  "PAN": "1111110000000003",
  "expirationDate": {
    "month": "11",
    "year": "2024"
  }
}
```

//Plain NewCardInfo Object Data:

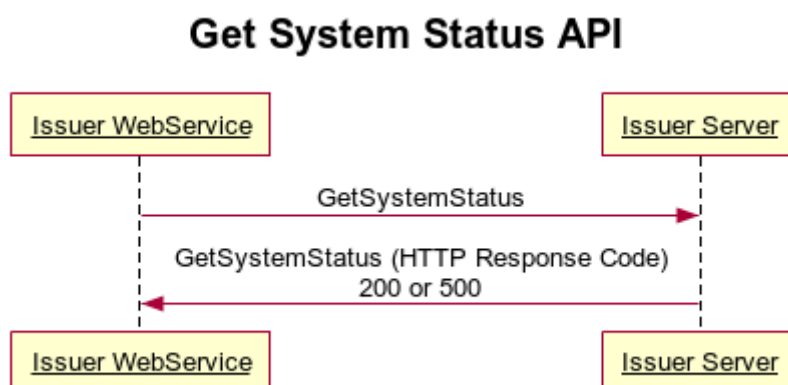
```
{
  "PAN": "1111110008484383",
  "expirationDate": {
    "month": "05",
    "year": "2026"
  }
}
```

ChangeCardInfoResponse

```
{
  "requestID": "4",
  "returnCode": "00"
}
```

5.7. GetSystemStatus

This API is used to check the system's health status.



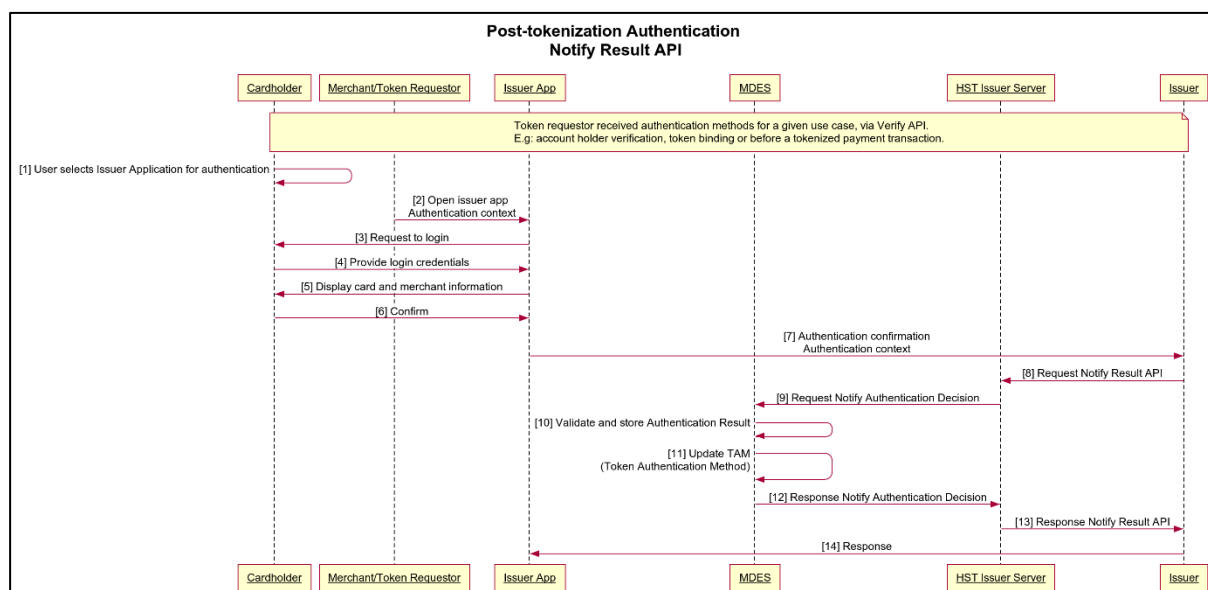
API endpoint	Method
Sandbox: https://issuer-bus.test-teste-prueba.com:9215/api/v3/getsystemstatus	GET
Production: https://issuer-bus.shieldedtransaction.com:9215/api/v3/getsystemstatus	GET

Issuer Server responds with 200 if OK or 5XX in case of error or unavailability.

5.8. NotifyResult

This API allows issuers to update token corresponding issuer decision after cardholder authentication.

For now, this API is only used for Mastercard implementations to support TAF – Token Authentication Framework, when cardholder selects Issuer App or Issuer Web Portal for authentication.

**API endpoint****Method**

Sandbox: <https://issuer-bus.test-teste-prueba.com:9215/api/v3/notifyresult>

POST

Production: <https://issuer-bus.shieldedtransaction.com:9215/api/v3/notifyresult>

POST

NotifyResultRequest

Element:	requestID
Description:	Request identifier uniquely generated for each request by the Issuer.
Type:	String
Size:	Max 64
Required:	Yes
Element:	institutionCode
Description:	A code generated by HST that identifies the Issuer during the request.
Type:	String
Size:	Max 32
Required:	Yes
Element:	apiVersion
Description:	Indicate the version of the API that must be used.
Type:	String
Size:	Max 16
Required:	Optional
Element:	serviceId
Description:	Unique identifier to choose the service needed. Options are: 1 – TK

Type:	2 – ACS
Size:	String
Required:	Max 2
Element:	Yes
Element:	Brand
Description:	Possible values are: 1 – Visa 2 – Mastercard 3 – Amex 4 – Elo 5 – Carnet
Type:	String
Size:	Max 2
Required:	Yes
Element:	tokenRefID
Description:	Token Reference ID associated to the token created to the specified card (EncryptedCardInfo) previously. It is a value assigned by the vault. Each token generates a tokenRefID value.
Type:	String
Size:	Max 64
Required:	Yes
Element:	appID
Description:	The issuer application or web portal identifier. Used for auditing purposes only.
Type:	String
Size:	4-50
Required:	Yes
Element:	appName
Description:	Issuer application or web portal name. Used for auditing purposes only.
Type:	String
Size:	4-50
Required:	Yes
Element:	organization
Description:	Name of the Issuer or Processor who triggered the API request. Used for auditing purposes only.
Type:	String
Size:	4-200
Required:	Yes
Element:	reason
Description:	The reason why the authentication was made. Used for auditing purposes only. For example: “Cardholder authentication performed post tokenization”
Type:	String
Size:	Max 254
Required:	Optional

Element:	result
Description:	Authentication result info.
Type:	NotificationResult Object
Required:	Yes

NotifyResultResponse

Element:	requestID
Description:	Request identifier unique generated for each request by HST.
Type:	String
Size:	Max 64
Required:	Yes
Element:	errorDescription
Description:	Error description returned only in error conditions for troubleshooting purposes.
Type:	String
Size:	Max 256
Required:	Optional
Element:	returnCode
Description:	0 – Success 89 – Duplicated request 97 – Required data not found 98 – Invalid request 99 – Internal error
Type:	String
Size:	Max 3
Required:	Yes

6. General Objects

6.1. CardMetaData

Element:	foregroundColor
Description:	Foreground color of the Digital Wallet entry for the card. (i.e. rgb(12,225,585))
Type:	String
Size:	0-32
Required:	Optional – Only available on VTS
Element:	backgroundColor
Description:	

Type:	Background color of the Digital Wallet entry for the card. (i.e. rgb(14,245,095))
Size:	String
Required:	0-32 Optional – Only available on VTS
Element:	labelColor
Description:	Label color of the Digital Wallet UI entry (“space”) for the card. (i.e. rgb(06,321,769))
Type:	String
Size:	0-32
Required:	Optional – Only available on VTS
Element:	shortDescription
Description:	A short description of the card.
Type:	String
Size:	0-32
Required:	Optional – Only available on VTS
Element:	longDescription
Description:	A long description of the card.
Type:	String
Size:	0-64
Required:	Optional – Only available on VTS
Element:	contactEmail
Description:	Customer Service’s e-mail of the issuer bank.
Type:	String
Size:	0-64
Required:	Optional – Only available on VTS
Element:	contactPhone
Description:	Customer Service’s phone number of the issuer bank.
Type:	String
Size:	0-32
Required:	Optional – Only available on VTS
Element:	contactName
Description:	Issuer bank’s name.
Type:	String
Size:	0-64
Required:	Optional – Only available on VTS
Element:	termsAndConditionsID
Description:	Issuer bank terms and conditions Id configured on the Vault.
Type:	String
Size:	0-32
Required:	Optional – Only available on VTS
Element:	cardArtID
Description:	Issuer bank card art Id configured on the Vault.
Type:	String
Size:	0-32

Required:	Optional – Only available on VTS
-----------	----------------------------------

Element:	productId
Description:	Unique identifier of the card product as registered on the platform.
Type:	String
Size:	0-32
Required:	Optional for “VTS” and “MDES” Required for “AMEX” Deprecated – New implementations should use “<i>productID</i>” instead

Element:	productID
Description:	Unique identifier of the card product as registered on the platform.
Type:	String
Size:	0-32
Required:	Optional for “VTS” and “MDES” Required for “AMEX”

Element:	productName
Description:	Card product name (description).
Type:	String
Size:	0-12
Required:	Required for “AMEX”

Element:	productType
Description:	Card type. For example: “CREDIT”, “DEBIT”, “PREPAID”.
Type:	String
Size:	0-64
Required:	Required for “AMEX”

6.2. TokenInfo

Element:	token
Description:	Token Value assigned to the PAN.
Type:	String
Size:	13-19
Required:	Optional

Element:	expirationDate
Description:	Card’s expiration date.
Type:	ExpirationDate
Required:	Optional

Element:	state
Description:	“ACTIVE”, “SUSPENDED”, “INACTIVE”, “CANCELED”.
Type:	String
Size:	Max 16
Required:	Optional

Element:	type
Description:	"HCE", "SE", "ECOM", "QRCODE", "COF".
Type:	String
Size:	Max 16
Required:	Optional Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".
Element:	lastTokenStatusUpdatedTimeStamp
Description:	Format: yyyy-MM-ddTHH:mm:ss.SSSZ The value will be in GMT.
Type:	String
Size:	Max 32
Required:	Optional
Element:	entityOfLastAction
Description:	"TOKEN_REQUESTOR" or "ISSUER".
Type:	String
Size:	Max 16
Required:	Optional
Element:	deviceInfo
Description:	It will not be present for tokens that are not device bound.
Type:	Object
Size:	1
Required:	Optional
Element:	OTPCodeIndicate
Description:	"PRESENT", "NOT_PRESENT" or "EXPIRED".
Type:	String
Size:	Max 16
Required:	Optional
Element:	OTPCodeExpiration
Description:	Format: YYYY-MM-DDThh:mm:ss.SSSZ The value will be in GMT.
Type:	String
Size:	Max 32
Required:	Optional
Element:	PANLastFour
Description:	These are the last four digits of the current PAN for the token.
Type:	String
Size:	4
Required:	Optional
Element:	previousPANLastFour
Description:	These are the last four digits of the previous PAN for the token. If a card has been replaced while the token was in an active state then this represents the previous PAN that the token was associated with.
Type:	String
Size:	4

Required:	Optional
Element:	tokenRefID
Description:	Identifier of the Token.
Type:	String
Size:	Max 64
Required:	Optional. Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".
Element:	activationFlow
Description:	Defines how the token was activated. Possible values: "GREEN" or "YELLOW".
Type:	String
Size:	Max 64
Required:	Optional
Element:	panSource
Description:	Indicates how the PAN was provided. Possible values are: "ON_FILE" – PAN origin is a card number stored in a merchant; "MANUALLY" – PAN was entered by the customer; "MOBILE_APP" – PAN provided by a mobile app. Typically a list of cards provided by the issuer after cardholder authentication; "TOKEN" – The source of pan of this token (ECOM o COF) provisioning was issued by a token device bound (NFC/SE). Applicable to a scenario such as a wallet has an NFC/SE token and it is provisioning a new E-Commerce/COF token. Used too when external account token was provided for migration (MDES Multi Device Provisioning).
Type:	String
Size:	Max 64
Required:	Optional. Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".
Element:	activationMethod
Description:	Describes how the token was activated: Possible values are: "AUTOMATIC" (green flow), "STEPUP_OTP", "STEPUP_CALL_CENTER", "STEPUP_ISSUER_APP", "UNKNOWN".
Type:	String
Size:	Max 64
Required:	Optional
Element:	activationDateTime
Description:	GMT Date and time of activation ("yyyy-MM-ddTHH:mm:ss.SSSZ")
Type:	String
Size:	64
Required:	Optional

Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".	
Element:	tokenAssuranceLevel
Description:	The assurance level assigned to the token.
Type:	String
Size:	2
Required:	Optional Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".
Element:	tokenRequestorID
Description:	Identification of the Token Requestor requesting digitization. It identifies SamsungPay, ApplePay, a Multi Issuer Wallet or an Issuer Wallet. All the Token Requestor ID values are generated by the brand and a table is provided by them to Issuers during the initial steps of the project. If provided, results will only contain tokens related to that specific Token Requestor ID.
Type:	String
Size:	Max 64
Required:	Optional
Element:	tokenRequestorName
Description:	Identification of the Token Requestor Name dynamically reported by the vault, it is present in DigitizationNotificationAPI payload. When provided updates the audit reports in Pay Admin.
Type:	String
Required:	Optional
Element:	deletedFromApp
Description:	It indicates if the token is deleted only from the device/token requestor or both device and the MDES platform. For Apple Pay tokens deleted from the device don't produce automatic notifications to the issuers.
Type:	Boolean
Required:	Optional – MDES Only (Apple Pay)
Element:	isoDeviceID
Description:	Two digit device type provided on the ISO messages that the source token has been provisioned. In Appendix B , we created a table with all possible values.
Type:	String
Required:	Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".

Element:	sourceToken
Description:	The source token account number used for the provisioning request.
Type:	String
Required:	Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".
Element:	provisioningScope
Description:	Indicates the provisioning context which is applicable to device wallets. Possible values: "ADP" - (Authorized Device Provisioning) The wallet authenticates the user on both devices before initiating the tokenization request. The wallet shares existing token information as account data; "MDP" - (Multi Device Provisioning) After provisioning to one device, the cardholder requests to push the same account to additional devices. The wallet shares existing token information as account data.
Type:	String
Required:	Conditional only to MDES when source element in CheckEligibility API is " TOKEN ".

6.3. DeviceInfo

Element:	deviceType
Description:	"UNKNOWN", "MOBILE_PHONE", "TABLET", "WATCH", "PC".
Type:	String
Size:	Max 32
Required:	Optional
Element:	deviceNumber
Description:	Mobile phone number or last four digits of mobile phone number.
Type:	String
Size:	Max 13
Required:	Optional
Element:	deviceName
Description:	User assigned device name.
Type:	String
Size:	Max 16
Required:	Optional
Element:	deviceModel
Description:	Model of the device.
Type:	String
Size:	Max 32
Required:	Optional
Element:	serialNumber
Description:	Masked Serial Number.
Type:	String
Size:	Max 32

Required:	Optional
Element:	deviceID
Description:	The unique device identifier.
Type:	String
Size:	Max 48
Required:	Optional
Element:	deviceIndex
Description:	The index number from Vault where deviceID is stored. Required for token device binding.
Type:	String
Size:	Max 2
Required:	Optional
Element:	deviceIPAddressV4
Description:	The IP address of the device at the time of the provisioning request, with the value represented in 255.255.255.255 format.
Type:	String
Size:	7-64
Required:	Optional
Element:	locationSource
Description:	Latitude and longitude where the device of the consumer is attempting to authorize is located. It can be obfuscated.
Type:	String
Size:	Max 64
Required:	Optional
Element:	osType
Description:	Type of operating system. It is one of the following values: "ANDROID", "IOS", "WINDOWS", "OTHER".
Type:	String
Size:	Max 32
Required:	Optional
Element:	osVersion
Description:	Version of the operating system running on the device.
Type:	String
Size:	Max 32
Required:	Optional
Element:	tokenProtectionMethod
Description:	Method of token protection. It is one of the following values: "SOFTWARE", "TRUSTED_EXECUTION_ENVIRONMENT", "SECURE_ELEMENT", "CLOUD", "OTHER".
Type:	String
Size:	Max 32
Required:	Optional
Element:	deviceLanguageCode
Description:	ISO 639 Version 3 Language Code value, for example, eng (English).
Type:	String

Size:	3
Required:	Optional
Element:	deviceManufacturer
Description:	Manufacturer of the device.
Type:	String
Size:	Max 32
Required:	Optional
Element:	deviceIMEI
Description:	Hardware ID of the device (IMEI).
Type:	String
Size:	Max 24
Required:	Optional

6.4. AuthenticationMethod

Element:	identifier
Description:	Required if cardholder verification method is returned. Identifies each verification method during the issuer response, which means is unique and opaque identifier for each method. This ID should be defined and provided by the issuer.
Type:	String
Size:	0-32
Required:	Yes
Element:	type
Description:	The available options are: “cell_phone” – OTP sent to cell phone number; “email” – OTP sent to e-mail address; “bank_app” – Authentication through the mobile issuer app (Android/iOS). Note: Set <i>sourceAddress</i> element value as “PUSH_NOTIFICATION” if the issuer wants to support application that opens through push notification along with Android intent and iOS deep link URL. <u>PUSH_NOTIFICATION value is not applicable for activating a device token, it is only applicable for post-tokenization authentication (Mastercard TAF);</u> “customer_service” – Authentication through issuer call center; “outbound_call” – Call received by the cardholder; “emv_3ds” - EMV 3DS will be used to authenticate the cardholder. <i>sourceAddress</i> value will be filled with 3DS URL; “bank_website” - Authentication through the web issuer app. Value will be the website URL (Issuer web app); “push_notification_to_mobile_app” - Push Notification to mobile app instance of cardholder <u>to receive an authentication code, it is</u>

Type:	only applicable for Post-tokenization authentication (Mastercard TAF).
Size:	String
Required:	Max 64
Element:	Yes
Element:	maskedInfo
Description:	Masked Consumer (cell phone): '*****19' Masked Consumer (email_address): 'ip****@gmail.com'. Mobile Banking (bank_app): 'Mobile Banking App' Call Center (customer_service): '1-800-555-555'
Type:	String
Size:	Max 64
Required:	Yes
Element:	customerAddress
Description:	email: 'testcustomer@gmail.com'. phone number: '1-800-555-555'
Type:	String
Size:	Max 64
Required:	Optional – only for auditing purpose
Element:	sourceAddress
Description:	When used with 'Type' 'bank_app', this value must contain the appropriate identifier for the associated issuer mobile banking application, such as " <i>com.DemoBank.DemoApp</i> " for example. For Apple this would be the Apple Adam ID and for Android this would be the Android Package name. When type is "emv_3ds", this element must be filled with 3DS URL.
Type:	String
Size:	Max 64
Required:	Optional – only used for Bank App flow
Element:	platform
Description:	This field is used when the type of field contains the value bank_app . Valid Values: "IOS", "ANDROID", "WINDOWS", "WEB".
Type:	String
Size:	Max 64
Required:	Optional

6.5. ExpirationDate

Element:	month
Description:	Month of expiry date.
Type:	String
Size:	2
Required:	Yes

Element:	year
Description:	Year of expiry date (i.e. XXXX).
Type:	String
Size:	4
Required:	Yes

6.6. CardInfo

Element:	PAN
Description:	Primary Account Value.
Type:	String
Size:	16-19
Required:	Yes
Element:	expirationDate
Description:	Card expiration date.
Type:	ExpirationDate
Required:	Optional - Required for Click to Pay
Element:	CVV2
Description:	Card Verification Value presented on the back of the physical card.
Type:	String
Size:	3
Required:	Optional
Element:	cardholderName
Description:	Cardholder Name as it appears on card. Special characters or numbers are not valid.
Type:	String
Size:	Max 60 for Click to Pay Max 32 for other tokenization uses
Required:	Optional - Required for Click to Pay
Element:	PANSequence
Description:	Funding account PAN sequence. Examples: 00 (Default Value), 01, 02, 03.
Type:	String
Size:	Up to 3
Required:	Required for "AMEX". Conditional for MDES – Present when <i>Tap to Add Card</i> feature is enabled and if chipdata is present in the encryption payload

6.7. EncryptedPayload

Element:	algorithm
Description:	

Type:	Encryption Algorithm used to protect data. Supported types are: “ aes-gcm128 ”, “ aes-ccm128 ”, “ aes-gcm256 ”, “ aes-ccm256 ”, “ jwe ”. Refer to https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38c.pdf and https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-38d.pdf
Required:	String Yes
Element:	nonce
Description:	Nonce for AES_CCM.
Type:	String
Size:	7, 8, 9, 10, 11, 12 or 13 (if not sure, 11 should be used)
Required:	Optional
Element:	iv
Description:	Initial Vector for AES_GCM. This field is a String which represents an array of 32 hexadecimal digits, representing at most 16 bytes.
Type:	String
Size:	32
Required:	Optional
Element:	encryptedData
Description:	Encrypt Data value using SIK . All the ciphered data must be transmitted in base64.
Type:	String
Size:	0-256k
Required:	Yes
Element:	associatedData
Description:	Data that is not encrypted but used for MAC calculation.
Type:	String
Size:	0-256K
Required:	Optional
Element:	MACLength
Description:	Specifies the MAC length that will be generated. The MAC contents are located at the end of the encryptedData element. Valid values for CCM algorithm : 4, 6, 8, 10, 12, 14 or 16 bytes (reasonable minimum is 12). Valid values for GCM algorithm : 4, 6, 8, 10, 12, 14 or 16 bytes (reasonable minimum is 12).
Type:	Numeric
Required:	Yes

6.8. TokenUserInfo

Element:	ID
Description:	The unique value that identifies the token user. (The entity which initiates the payment request).

Type:	String
Size:	11
Required:	Yes
Element:	appType
Description:	Application type for the token user. This entity can be the merchant, a marketplace, or a checkout host. Possible values are: "WEB" "MOBILE_APP" "MOBILE_WEB" "MARKETPLACE" "VOICE_APP" "BIOMETRIC_APP"
Type:	String
Required:	Optional

6.9. MerchantInfo

Element:	ID
Description:	The unique value that identifies the merchant. Required for trusted listing enrollment.
Type:	String
Size:	8
Required:	Optional
Element:	merchantName
Description:	The merchant name.
Type:	String
Size:	256
Required:	Optional

6.10. RiskInformation

Element:	riskAnalysisServiceProvider
Description:	Indicates who is the service provider selected to analyze the transaction risk. Options are: 1 – HST 2 – Mastercard 3 – Visa
Type:	String
Size:	3
Required:	Optional
Element:	recommendedDecision
Description:	

Type:	The decision recommended by the Wallet Provider (token requestor). Possible values are: "GREEN", "YELLOW", "ORANGE" or "RED".
Size:	String
Required:	64 Optional
Element:	deviceScore
Description:	Score given to the device by the Wallet Provider (token requestor). Value between 1 and 5, where 5 indicates the most confidence on the device.
Type:	String
Size:	2
Required:	Optional
Element:	accountScore
Description:	Score given to the account by the Wallet Provider (token requestor). Value between 1 and 5, where 5 indicates the most confidence on the account.
Type:	String
Size:	2
Required:	Optional
Element:	vaultRiskAssessmentScore
Description:	Advanced Authorization (AA) Risk Score is generated by VisaNet on the last payment transaction for the PAN. Values are 00 through 99, with a higher value indicating higher risk. This value is calculated by the AA Risk engine, based on the PAN's transaction pattern (long-term and short-term).
Type:	String
Size:	2
Required:	Optional for VTS. Not present otherwise.
Element:	vaultTokenScore
Description:	<u>VISA</u> - Value indicating the degree of risk associated with the token. Numeric value is 00 through 99. A score of 00 indicates that card brand did not provide a score.
	<u>MASTER</u> – Value from 0-9 and represents the Token Provisioning Score (TPI). The lowest risk is 0, the highest risk is 9.
Type:	String
Size:	Max 2
Required:	Optional for Visa and Mastercard, not present otherwise.
Element:	phoneNumberScore
Description:	Score given to the phone number by the Wallet Provider (token requestor). Value between 1 and 5, where 5 indicates the most confidence on the phone number.
Type:	String
Size:	2

Required:	Optional
Element:	provisioningAttemptsOnDeviceIn24Hours
Description:	Number of provisioning attempts on this device in the past 24 hours. If the number of attempts exceeds 99, it will stay at 99.
Type:	String
Size:	2
Required:	Optional
Element:	accountLifeTime
Description:	The lifetime of the account with the Token Requestor. Value is in months.
Type:	String
Size:	2
Required:	Optional
Element:	recommendationReasons
Description:	Reasons provided to the Token Requestor on how the recommended decision was reached.
Type:	Array<String>
Required:	Optional
Element:	securityReasonCode
Description:	Values from AA-ZZ representing reason codes sent by Token Provisioning Insights (TPI). AA is a higher risk and ZZ is a lower risk reason. Review the AppendixA for further information.
Type:	String
Size:	2
Required:	Optional – Used only by Mastercard.

6.11. TermsAndConditions

Element:	id
Description:	The terms and conditions identifier generated by the Vault.
Type:	String
Size:	64
Required:	Optional
Element:	date
Description:	The date and time the terms and conditions were accepted by the cardholder. Format: YYYY-MM-DDThh:mm:ss.SSSZ The value will be in GMT.
Type:	String
Size:	0-32
Required:	Optional

6.12. Market

Element:	countryCode
Description:	Two letter country code based on ISO 3166. Example: "BR", "US", "MX".
Type:	String
Size:	2
Required:	Required for "AMEX".
Element:	regionName
Description:	Region name is the country's name.
Type:	String
Size:	0-64
Required:	Required for "AMEX".
Element:	locale
Description:	Locale in xx_XX format. The format is based on xx_XX, where xx refers to Language code and XX refers to Country code. Examples: en_US, en_SA, pt_BR, es_MX, etc. Note: ISO standard values for the country of the Issuer.
Type:	String
Size:	0-12
Required:	Required for "AMEX".

6.13. ReturnObject

Element:	returnCode
Description:	The error code associated to the API business error. "00" – OK.
Type:	String
Size:	16
Required:	Required
Element:	errorDetails
Description:	The error details associated to the API business error.
Type:	ErrorObject
Required:	Optional

6.14. ErrorObject

Element:	cardID
Description:	Card identification provided by the issuer. This will be used as return to the issuer when receiving the bulk provision results.
Type:	String
Size:	0-36
Required:	Conditional – This field is required in the GetBulkJobStatus response, specifically in the errorCardList element.
Element:	errorCode
Description:	The error code is associated with the API business error.

Type:	String
Size:	16
Required:	Required
Element:	errorDescription
Description:	The error description is associated with the API business error.
Type:	String
Size:	256
Required:	Required
Element:	encryptedCardProfileListIndex
Description:	Indicates the position of the record in the request that was processed with error. Used in scenarios where a cardID cannot be assigned to the record.
Type:	Numeric
Required:	Conditional

6.15. ChipData

Element:	iccSystemRelatedData
Description:	The Account Primary Account Number of the card to be digitized. It contains Hex-encoded string.
Type:	String
Size:	Up to 255
Element:	track2Data
Description:	The track2Data of the card to be digitized. It contains Hex-encoded string.
Type:	String
Size:	Up to 37
Element:	panEntryMode
Description:	The method used for PAN entry to initiate the digitization. One of the possible values: “05” = Automatic entry of the PAN into the merchant terminal through an integrated circuit card (reserved for the DIP Card). “07” = Automatic entry of the PAN into the merchant terminal via Contactless M/Chip.
Type:	String
Size:	Up to 4

6.16. ChipDataValidationResult

Element:	OBSServiceIndicator
Description:	The On-behalf Service indicator of the chip Data validation. One of the possible values: “02” = Pre-validation Service of the M/Chip Cryptogram “03” = Validation of the M/Chip Cryptogram in Stand-In Processing
Type:	String
Size:	Up to 2

Element:	OBSResult
Description:	The On-behalf Service chip Data validation result. One of the possible values is valid: "A" = Valid Application Cryptogram (AC); ATC out of permitted range "F" = Format Error "K" = Key file does not match for this combination of PAN, PAN expiration date and KDI "U" = Unable to process "V" = Valid "X" = Security platform interruption "Z" = Security platform processing error
Type:	String
Size:	Up to 3

6.17. PaymentData

Element:	transactionID
Description:	Unique identification of transaction shared by token requestor. Example: "3232323".
Type:	String
Size:	Up to 32
Required:	Yes
Element:	transactionAmount
Description:	Transaction amount. For example: "1234".
Type:	String
Size:	Up to 48
Required:	Yes
Element:	currencyCode
Description:	Transaction currency code ISO format. For example: "BRL", "USD".
Type:	String
Size:	3
Required:	Yes

6.18. TKData

Element:	vaultIdentification
Description:	Possible values are: "VTS" – for Visa; "MDES" – for Mastercard; "AMEX" – for Amex;

Type:	"PL" – for Private Label.
Size:	Used to identify the Vault in case tokenRefID does not exist in HST database.
Required:	String
	32
	Yes
Element:	PANRefID
Description:	The PANRefID is a value assigned by the vault to identify the PAN. It identifies the PAN on the Vault. For VISA, each PAN generates a PANRefID value, which means a VISA PAN must have only one PANRefID value assigned. For Mastercard, the PANRefID is associated to the Token Requestor, which means it is not unique for a PAN and it can have multiples PANRefIDs. By using this data, it is not necessary to input the real PAN value. It is recommended to relate the PANRefID value with the TokenRefID for further use in APIs such as GetAssociatedTokens, GetPANByPANRefID and others.
Type:	String
Size:	64
Required:	Required for "VTS" and "MDES" Not present for "AMEX"
Element:	authenticationRequestID
Description:	The Identifier value to link between different authentication requests/messages.
Type:	String
Size:	64
Required:	Yes
Element:	reasonCodes
Description:	The possible values are: "VERIFY_ACCOUNT" - The account holder is being authenticated to verify account ownership. "PAYMENT" - the account holder is being authenticated to initiate payment transaction. "TOKEN_BINDING" - The account holder is being authenticated to bind the token with bind id. "ACCOUNT_BINDING" - The account holder is being authenticated to bind the card number which is tokenized with bindID. "OTHER" - The account holder is being authenticated for a reason not enumerated in this list.
Type:	Array <String>
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework) and SCOF post tokenization.

Element:	bindID
Description:	The unique and fixed identifier to a component of data bound to the token credential.
Type:	String
Size:	48
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when reasonCode value is "TOKEN_BINDING" or "ACCOUNT_BINDING".
Element:	authenticatorInfo
Description:	Contains authenticator's information that authenticates cardholders for subsequent payment transactions.
Type:	AuthenticatorInfo Object.
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when reasonCode value is "TOKEN_BINDING" or "ACCOUNT_BINDING".
Element:	recentAuthenticationInfo
Description:	Cardholder recently authenticated by this entity. Applicable only for post-tokenization authentication.
Type:	RecentAuthenticationInfo Object.
Required:	Optional
Element:	certifiedMFAAuthMethodId
Description:	Card brand certified multi-factor authentication method ID that authenticator will use to authenticate cardholder for subsequent authentication after token or Account has been bound with given bind ID.
Type:	String
Size:	1-5
Required:	Conditional. Used for Mastercard TAF (Token Authentication Framework), when reasonCode value is "TOKEN_BINDING" or "ACCOUNT_BINDING".

6.19. AuthenticatorInfo

Element:	entityID
Description:	Identification of authenticating entity.
Type:	String
Size:	Up to 5
Required:	Yes
Element:	entityName
Description:	Name of authenticator.
Type:	String
Size:	Up to 32
Required:	Optional

6.20. RecentAuthenticationInfo

Element:	trustedAuthorityName
Description:	Name of card brand trusted authenticator who recently authenticated cardholder.
Type:	String
Size:	Max 64
Required:	Optional
Element:	trustedAuthorityID
Description:	Identification of card brand trusted authenticator who recently authenticated. Example: "654".
Type:	String
Size:	Max 5
Required:	Optional
Element:	authenticationProof
Description:	Authentication proof shared by the trusted authenticator. For example: "OTP".
Type:	String
Size:	35
Required:	Optional
Element:	authenticationTime
Description:	Date and time when this authentication happened. Expressed in ISO 8601 extended format as one of the following: - YYYY-MM-DDThh:mm:ss[.sss]Z, YYYY-MM-DDThh:mm:ss[.sss]±hh:mm, where [.sss] is optional and can be 1 to 3 digits.
Type:	String
Size:	Max 32
Required:	Yes

6.21. NotificationResult

Element:	vaultIdentification
Description:	Possible value currently is: "MDES" – for Mastercard.
Type:	String
Size:	32
Required:	Yes
Element:	authenticationRequestID
Description:	Unique identifier for the authentication request.
Type:	String
Size:	Max 64
Required:	Yes

Element:	authenticationMethod
Description:	Method used to authenticate. Options are: 7 – "bank_app" 11 – "bank_website"
Type:	String
Size:	Max 2
Required:	Yes
Element:	resultStatus
Description:	Authentication/challenge result. Options are: 9 – Challenge confirmed/Successful authentication; 10 – Challenge denied/Failed authentication; 11 – Challenge expired;
Type:	String
Size:	Max 2
Required:	Yes

6.22. ComplementaryAuthenticationInfo

Element:	CDCVM
Description:	Consumer device cardholder verification method. One of the following values is valid: "UNKNOWN", "NONE", "SIGNATURE", "ONLINE_PIN", "PASSCODE", "DEVICE_CODE", "BIOMETRIC", "DEVICE_PATTERN".
Type:	String
Size:	Up to 32
Required:	Optional
Element:	tokenVerificationResultCode
Description:	Result of token verification. One of the following values is valid: "TAVV_FAILED", "TAVV_PASSED", "DTVV_FAILED", "DTVV_PASSED", "AUTHORIZED_DEVICE_TRANSFER_PASSED".
Type:	String
Size:	Up to 64
Required:	Optional

7. Return Codes

Code	Description
00	Ok
05	Card not eligible
11	Invalid Institution Code
13	Too many items enqueued to be processed, try again later
16	Card not found, invalid PAN
22	Invalid Card Security Code
23	Invalid Card Expiration Date
24	Card has not been activated, replaced, or renewed card has not been activated
25	Non-whitelisted accounts when a market is at beta test phase
26	Ineligible instant account/instant membership account provisioning
27	Too many attempts, suspected fraud. Return expected when element "recommendedDecisionReasonCode" value received in Check Eligibility request is "0002"
85	Requires ID&V
89	Duplicated request
90	Connection Timeout Error
91	Invalid Token Status/Token Not Active
92	Token Not Found
93	Token Already in the State Requested
94	Invalid Replacement PAN
95	Cryptography Error
96	Invalid Data
97	Required Data Missing
98	Invalid Request
99	System Error

8. Appendices and Supplementary Information

8.1. Appendix A

Mastercard Token Provisioning Insights - Reason Code Table

This data is received from MDES to support issuers in performing a detailed risk assessment for each pre-provisioning token request.

The data is delivered to issuers by HST via the securityReasonCode element, which is included within the riskInformation object of the CheckEligibility request.

Reason Code	Description
AB	Suspicious transaction linked to card.
AG	Previous cross border transactions may signal risk.
AY	High-velocity provisioning activity by token requestor.
BT	Suspicious token requestor behavior.
HW	High transaction volume prior to provisioning.
HZ	Suspicious provisioning request due to multiple risk indicators.
JA	Unable to generate provisioning insight. A Reason Code of JA will be returned if a score is unable to be generated, and the Risk Score will automatically be set to 5 .
LL	New card with no recent provisioning activity.
NR	No recent Token Requestor activity.
SG	Provisioning behavior appears normal and stable.
ST	Normal Token Requestor behavior.
TC	Provisioning Request from banking app.
YH	New token request from a low-risk token requestor and consistent activity with this card.

8.2. Appendix B

Mastercard ISO Device Type Table

The two digit device type provided on the ISO messages that the source token has been provisioned.

In this table are all available and possible values:

Code	Device Type
00	Card
01	Mobile Network Operator (MNO) controlled removable secure element (SIM or UICC) personalized for use with a mobile phone or smartphone
02	Key fob
03	Watch using a contactless chip or a fixed (non-removable) secure element not controlled by the MNO
04	Mobile tag
05	Wristband
06	Mobile phone case or sleeve
07	Mobile phone or smartphone with a fixed (non-removable) secure element controlled by the MNO, for example, code division multiple access (CDMA)
08	Removable secure element not controlled by the MNO, for example, memory card personalized for used with a mobile phone or smartphone
09	Mobile phone or smartphone with a fixed (non-removable) secure element not controlled by the mobile network operator
10	MNO controlled removable secure element (SIM or UICC) personalized for use with a tablet or e-book
11	Tablet or e-book with a fixed (non-removable) secure element controlled by the MNO
12	Removable secure element not controlled by the MNO, for example, memory card personalized for use with a tablet or e-book
13	Tablet or e-reader with a fixed (non-removable) secure element not controlled by the mobile network operator
14	Mobile phone or smartphone with a payment application running in a host processor
15	Tablet or e-book with a payment application running in a host processor
16	Mobile phone or smartphone with a payment application running in the TEE of a host processor
17	Tablet or e-book with a payment application running in the TEE of a host processor
18	Watch with a payment application running in the TEE of a host processor
19	Watch with a payment application running in a host processor
21	Phone - mobile phone
22	Tablet
23	Watch
24	Sticker
25	Personal computer
26	Device peripheral
27	Tag
28	Jewelry
29	Fashion accessory
30	Garment

31	Domestic appliance
32	Vehicle
33	Media or gaming device
34	Virtual reality headset
99	None of above

9. Revision History

Date	Version	Description	Author
06/28/2024	3.8.3 (24.06)	- Add a new value "CONTACTLESS_TAP" in Element <i>Source</i> at the CheckEligibility API request - Add a new value "PC" in Element <i>DeviceType</i> at section 6.3 <i>Device Info</i>. - Removed duplicated value "TABLET" in Element <i>DeviceType</i> at section 6.3 <i>Device Info</i>. - Adjusted the name of element isTsAndCSAccepted to isTsAndCsAccepted at section 6.15 (<i>Provider Object</i>). - Elements <i>tokenRefID</i> and <i>deviceID</i> on Section 6.16 <i>ClientInformation</i> are now optional. - Changed the size of element <i>bulkPushReceiptID</i> on the <i>GetBulkJobStatus</i> API Request to 36.	Daniel P. Santo
10/30/2024	3.8.4 (24.10)	- Adjusted broken hyperlinks. - New elements <i>chipData</i> and <i>chipDataValidationResult</i> were included in Check Eligibility request. - New Object 6.20 <i>ChipData</i> was added in General Objects. - New Object 6.21 <i>ChipDataValidationResult</i> was added in General Objects. - Element <i>PANSequence</i> in <i>CardInfo</i> object was changed. - New element <i>chipDataValidationResponse</i> was added in Check Eligibility Response - Element <i>Source</i> in <i>DigitizationNotification</i> request API was changed to support a new value ("CONTACTLESS_TAP"). - In <i>ActivateToken</i> API request, <i>tokenRequestorID</i> field is now required.	Daniel P. Santo, José A. Ramos
12/12/2024	3.8.5 (24.12) Alpha	Description of element <i>processID</i> was changed in the request of the following APIs: Check Eligibility, Digitization Notification, SendPasscode and Lifecycle Notification.	José Antonio Ramos
01/22/2025	3.8.6 (25.01)	Element <i>externalConsumerID</i> was included in the field <i>ClientInformation</i> , used in <i>PushCard</i> and <i>InitiateBulkJob</i> APIs	Mariana R. F. Sampaio

05/16/2024	3.8.7 (25.05)	- Added PAN replacement instructions for Visa cards to the <i>ChangeCardInfo</i> API; - Added item and institution limits to the <i>InitiateBulkJob</i> API request; - Corrected the object name <i>riskInformation</i> in the <i>CheckElegibility</i> API request and the <i>GetTokenInfo</i> API response; - Added a size attribute to the <i>cardID</i> element in the <i>cardProfile</i> object of the <i>PushCard</i> API request; - Added a format to the <i>contactPhone</i> element in the <i>ClientInformation</i> object, used in the following APIs request: <i>PushProvisioningNotification</i>, <i>PushCard</i> and <i>InitiateBulkJob</i>; - The <i>cardHolderName</i> element in the <i>CardInfo</i> object is now mandatory for Click to Pay cases in the <i>InitiateBulkJob</i> API request. - Added support for the JWE algorithm in the <i>EncryptedPayload</i> object, used in <i>InitiateBulkJob</i> API request. - A new return code '13' has been added to the <i>InitiateBulkJob</i> API response and documented in the 'Return Codes' list. - The <i>operation</i> field in the <i>ChangeCardInfo</i> API request was updated: the DELETE operation is no longer available.	Mariana R. F. Sampaio
07/04/2025	3.8.8 (25.07)	- The <i>locale</i> element in the <i>ClientInformation</i> object is now required in the <i>PushCard</i> API request; - Changed the Sandbox URLs for the <i>InitiateBulkJob</i> and <i>GetBulkJobStatus</i> APIs; - The <i>expirationDate</i> element is now mandatory in the 'CardInfo' object for Click to Pay requests in the <i>InitiateBulkJob</i> API; - Added a size attribute for the <i>bulkPushReceiptID</i> element in the <i>BulkProvisionNotification</i> API request and <i>InitiateBulkJob</i> API response. - Added a new optional element in <i>Check Eligibility Request</i> – “deviceType”.	Mariana R. F. Sampaio

		- Added a new optional element in Digitization Notification Request API – “tokenRequestorName”.	
11/10/2025	3.8.9 (25.11)	- A reference table specifying mandatory address fields per country has been incorporated for Mastercard requests at the end of section 6.14. - An error in the Intent data was identified and resolved by eliminating an unintended whitespace character, in section 6.15.	Leonardo Pereira
11/12/2025	3.9.0 (25.11)	- The element encryptedCardProfileListIndex was added to the ErrorObject structure on session 6.19, which is present in the GetBulkJobStatus API response; - Updated the standInReasonCode element options list in the DigitizationNotification API request; - Added a size parameter to the reason element in the ActivateToken API request, and clarified that this field is not present for Amex requests; - The cardID element is no longer mandatory in the errorObject included in GetBulkJob API request; - The state element in the BillingAddress object in the cardProfile object (used in InitiateBulkJob and PushCard API requests) now includes a size parameter; - The return element in the InitiateBulkJob API response has been renamed to returnObject; - The start element in the GetBulkJobStatus API response is now named startAt; - Optional DeviceInfo object was added to Check Eligibility API request; - New optional elements were added to DeviceInfo Object; - New optional elements were added to RiskInformation Object; - Updated DeviceInfo object with new elements: deviceIPAddressV4, locationSource, osType,	José Antonio Ramos Leonardo Pereira

osVersion, tokenProtectionMethod, deviceLanguageCode, deviceManufacturer and deviceIMEI;

- Updated RiskInformation object with new elements: phoneNumberScore and provisioningAttemptsOnDeviceIn24Hours;
- Added virtualCardRelatedData and merchantAdviceCode elements in CheckEligibility API response, regarding Mastercard Virtual Card Number mandate (**GLB 11165.1** announcement);
- Updated event element with PAYMENT_ACCOUNT_REFERENCE_UPDATED option, regarding Mastercard Virtual Card Number mandate (**GLB 11165.1** announcement).

12/04/2025	3.9.1 (25.12)	Updated Bulk packages limits per institution in section 5.9.1.	José Antonio Ramos Leonardo Pereira
12/05/2026	3.9.2 (25.12)	Updated intent and clientID objects with Mastercard specific options, in section 6.15.	José Antonio Ramos Leonardo Pereira
01/19/2026	3.9.3 (26.01)	Added a new size for object cardholderName in section 6.6.	José Antonio Ramos Leonardo Pereira
Adjusted TOKEN description on Source object from Check Eligibility request in section 4.1.			
05/02/2026	3.10.0 (26.02)	The Bulk Provisioning APIs have been relocated from this document to a new one, in which large-scale operations are centralized. (Bulk Enrollment Service [Click to Pay, Alias Directory Service])	José Antonio Ramos Leonardo Pereira
Mastercard TAF changes were inserted into sections listed below.			
Created: 4.8 Verify			
5.8 NotifyResult			
6.17 PaymentData			
6.18 TKData			

6.19 AuthenticatorInfor 6.20 RecentAuthenticationInfo 6.21NotifcationResult Modified: 4.3 SendPassCode 4.4 LifecycleNotification 6.2 TokenInfo 6.4 AuthenticationMethod 6.10 RiskInformation			
03/23/2026	3.11.0 (26.03)	Updated <i>vaultTokenScore</i> element inside of <i>riskInformation</i> object to receive risk score values from Mastercard TPI, other uses still the same. Created <i>securityReasonCode</i> element inside of <i>riskInformation</i> object to receive reason code values from Mastercard TPI. Created Appendix A in session 8.1, for further information about Mastercard TPI reason codes. Created <i>operationReasonCode</i> element inside <i>ChangeCardInfo</i> request to allow issuers to inform Visa the VAU operation reason. Updated <i>operation</i> element inside <i>ChangeCardInfo</i> to support the “DELETE” option. Updated the “Attention Required” notes at the end of <i>ChangeCardInfo</i> API with data needed for each VAU operation.	José Antonio Ramos Leonardo Pereira
06/03/2026	3.12.0 (26.06)	- Added object <i>ComplementaryAuthenticationInfo</i> , used in Apple Multi Device Provisioning scenario. - APIs <i>CheckEligibility</i> , <i>DigitizationNotification</i> and <i>LifeCycleNotification</i> were changed to add the optional object <i>ComplementaryAuthenticationInfo</i> used in Apple Multi Device Provisioning scenario.	José Antonio Ramos Leonardo Pereira Victor Nascimento

-
- Added element encryptedTokenInfo in CheckEligibility Request used in Apple Multi Device Provisioning scenario, session 4.1.
 - Added element provisioningDenyReason in CheckEligibility Response used in Apple Multi Device Provisioning scenario, at session 4.1.
 - Updated requirements for some elements inside tokenInfo object used in Apple Multi Device Provisioning scenario. Elements listed:
 - type
 - tokenRefID
 - panSource
 - activationDateTime
 - tokenAssuranceLevel
 - Updated description for TOKEN value from panSource element inside of tokenInfo object used in Apple Multi Device Provisioning scenario, session 6.2.
 - Created new elements inside tokenInfo object used in Apple Multi Device Provisioning scenario. Elements listed:
 - isoDeviceID
 - sourceToken
 - provisioningScope
-